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Numerical Exploration of Rectangular Eigenvalue Problems

Author:
Siyuan CHENG

Supervisor:
Prof. Daniel KRESSNER
Nian SHAO

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Abstract

Eigenvalue problems for square matrices are among the most fundamental tasks in numerical linear algebra, and a rich set of tools is available for their solution. When the matrix-valued function is rectangular, with more rows than columns, the determinant is undefined and exact eigenpairs generically do not exist. Two alternative characterizations guide the numerical treatment instead: residual minimization through the smallest singular value, and the minimum perturbation framework for rectangular linear pencils that seeks the nearest pencil admitting a prescribed eigenstructure. Both have been studied with various numerical methods. For nonlinear rectangular problems, the landscape is considerably sparser as most algebraic tools developed for the square nonlinear case do not extend to the rectangular setting.

This thesis assembles and tests a computational approach for nonlinear rectangular eigenvalue problems, combining rational approximation via sketchAAA, companion linearization, and minimum perturbation extraction via total least squares. We also introduce a cheap eigenvector recovery step that reuses the block structure of the companion pencil, and a cluster-aware post-processing step for the scan-refinement method that resolves closely spaced eigenvalues without refining the global grid.

Three numerical experiments evaluate this approach and its individual stages. For a Laplace eigenvalue problem on polygons, direct scan-refinement proves reliable, but the rational approximation route produces large and ill-conditioned companion pencils from which the minimum perturbation method fails to extract accurate eigenvalues. A quadratic rectangular problem arising from a damped wave equation removes the rational approximation stage from the pipeline and shows that, in this setting, the minimum perturbation extraction can recover accurate eigenvalues. A third experiment examines whether randomized sketching can reduce the cost of minimum perturbation extraction. The perturbation analysis demonstrate that sketching preserves the target eigenvector direction but can strongly distort the trailing subspace on which accurate extraction depends.

Contents

1 Introduction

Eigenvalue problems for square matrices are among the most fundamental tasks in numerical linear algebra. They can be written in the unified form

$$A(\lambda)x = 0,$$

where $A(\cdot) : D \subset \mathbb{C} \to \mathbb{C}^{n\times n}$ is a holomorphic matrix-valued function defined on a nonempty domain D in the complex plane and $x \in \mathbb{C}^n \setminus \{0\}$. When $A(\cdot)$ is regular, meaning that $\det(A(\lambda))$ is not identically zero on D, the eigenvalues are characterized by $\det A(\lambda) = 0$, reducing the problem to finding the zeros of a scalar analytic function. This framework includes the standard eigenvalue problem $A(\lambda) = A_0 - \lambda I$, where the determinant yields the characteristic polynomial, as well as the generalized eigenvalue problem $A(\lambda) = M - \lambda B$, where a regular pencil is commonly solved by the QZ algorithm [32]. For nonlinear eigenvalue problems with more general dependence on λ, standard approaches include Newton-type iterations [15], contour-integral methods [21, 4, 7, 17], and rational approximation with companion linearizations [21, 16, 20].

A rectangular eigenvalue problem breaks this algebraic framework. Here, one considers a rectangular matrix-valued function $A(\cdot) : D \to \mathbb{C}^{m\times n}$ with $m > n$, and the goal remains finding eigenpairs (λ, x) with $\lambda \in D$ and $x \in \mathbb{C}^n \setminus \{0\}$ satisfying $A(\lambda)x = 0$. Since $A(\lambda)$ is not square, the determinant is undefined, and exact eigenpairs generically do not exist: for most values of λ the matrix $A(\lambda)$ has full column rank. One natural perspective is to view the rectangular problem as a singular square problem. Indeed, appending $m - n$ zero columns to $A(\lambda)$ gives an $m \times m$ matrix-valued function with determinant identically zero, and in the linear case $A(\lambda) = M - \lambda B$ this corresponds to a singular pencil whose Kronecker canonical form contains a singular part [12].

With exact root-finding no longer applicable, two complementary characterizations guide the numerical treatment. First, approximate eigenvalues can be defined as local minimizers of the smallest singular value function

$$\sigma_{\min}(A(\lambda)) = \min_{\|x\|_2=1} \|A(\lambda)x\|_2,$$

and exact eigenvalues correspond to its zeros. This residual minimization viewpoint applies to any rectangular matrix-valued function. Second, for the important special case of linear pencils $A(\lambda) = M - \lambda B$, the minimum perturbation framework seeks the nearest perturbed pencil that admits a prescribed eigenstructure. Both formulations reduce to the determinant-based definition when $m = n$.

Rectangular eigenvalue problems arise naturally across a diverse range of computational and applied settings where systems are inherently overdetermined. In the numerical solution of differential equations, oversampled collocation methods are frequently used to suppress spurious solutions. For example, when extracting resonance frequencies, imposing more collocation conditions than basis functions [23], or stacking boundary and interior conditions in the method of particular solutions [3] and the method of fundamental solutions [1], deliberately construct a parameter-dependent rectangular eigenvalue problem. Beyond PDE collocation, the "distance to uncontrollability" of descriptor systems is mathematically characterized by the minimum perturbation required for a rectangular pencil to drop rank [8]. Furthermore, in data-driven fields such as signal processing, shape reconstruction from moments yields overdetermined pencils whenever the number of observed moments exceeds the theoretical unknown parameters [6]. Given this range of applications, the question of how to solve the rectangular eigenvalue problem efficiently and reliably has attracted several distinct approaches.

1.1 Existing approaches

1.1.1 Rectangular linear pencils

For rectangular linear pencils $A(\lambda) = M - \lambda B$ with $M, B \in \mathbb{C}^{m\times n}$ and $m > n$, several approaches have been studied. It is worth noting that one can always compress the pencil to a square one by a QR-based projection [23], but such reductions risk losing structural information. The methods below retain the rectangular geometry.

Minimum perturbation method. For a prescribed integer $1 \le \ell \le n$, one searches for the smallest matrix perturbations ΔM and ΔB measured in the Frobenius norm such that the perturbed pair $(M + \Delta M,\ B + \Delta B)$ admits exactly ℓ distinct eigenvalues with ℓ linearly independent eigenvectors. Boutry et al. [6] studied this problem for the single-eigenpair case $\ell = 1$ and developed a dedicated numerical algorithm. Chu and Golub [8] then provided a unifying framework for general ℓ. Their key insight is to shift the search from the matrix perturbations $(\Delta M, \Delta B)$ to the eigenvector subspace alone, optimizing over the Stiefel manifold $\{V_\ell \in \mathbb{C}^{n\times\ell} : V_\ell^{\mathsf{H}} V_\ell = I\}$. Their result gives the formula

$$\mu_\ell = \min_{V_\ell^{\mathsf{H}} V_\ell = I} \sqrt{\sum_{i=\ell+1}^{\min\{m,2\ell\}} \sigma_i^2([MV_\ell,\ BV_\ell])},$$

where μ_ℓ is the Frobenius norm of the smallest perturbation that forces ℓ eigenpairs to exist, the minimizer V_ℓ is an orthonormal basis for the ℓ-dimensional subspace spanned by these eigenvectors. While this formulation provides a profound theoretical reduction, it requires specialized numerical solvers. Li et al. [28] reformulated the task as an optimization problem over the product of two complex Stiefel manifolds, solving it via a Riemannian nonlinear conjugate gradient method. For the special case $\ell = n$ (a full set of eigenpairs), Ito and Murota [25] connected the framework to multidimensional total least squares (TLS), an approach that will be discussed in detail in Section 3.1.

Direct residual minimization. A third approach implements the residual minimization directly. Das and Neumaier [10] propose to interpret the eigenvalues of an overdetermined pencil $A - \lambda B$ as the local minimizers of

$$f(\lambda, v) = \frac{\|(A - \lambda B)v\|_2^2}{\|v\|_2^2}, \qquad v \neq 0,$$

or equivalently as the local minimizers of $\sigma_{\min}(A - \lambda B)$. For computation, they perform a two-sided orthogonal reduction of the pencil. Let $B = U_B R_B$ be a thin QR factorization of B. Left-multiplying the pencil by U_B^{H} extracts the $n \times n$ square part $R_0 - \lambda R$, where $R_0 = U_B^{\mathsf{H}} M$ and $R = R_B$. The remaining $m - n$ rows contribute a Hermitian positive semidefinite correction C that accounts for the rectangular excess. The reduced objective becomes

$$f_r(v, \lambda) = \frac{\|(R_0 - \lambda R)v\|_2^2 + v^{\mathsf{H}} C v}{\|v\|_2^2},$$

where R_0 and R are upper triangular and C is Hermitian positive semidefinite. The eigenpairs of the reduced square pencil $R_0 v = \lambda R v$ provide initial approximations, which are then refined by iterative updates of v and λ. In particular, their joint optimization scheme updates the eigenvector and eigenvalue simultaneously and is designed to locate the local minima of the reduced residual objective.

1.1.2 Rectangular nonlinear eigenvalue problems

The methods above all assume a linear pencil structure. For a general rectangular nonlinear $A(\lambda) \in \mathbb{C}^{m \times n}$, the landscape is considerably sparser. This is because most of the algebraic tools that drive the square nonlinear case [21, 4, 15, 19, 16] rely on a square determinant condition that does not extend to the rectangular setting.

Following the residual minimization characterization, one can scan $\lambda \mapsto \sigma_{\min}(A(\lambda))$ over a grid of trial values, identify local minima as eigenvalue candidates, and refine them by a bounded local search. This scan-refinement strategy is the basis of the method of particular solutions (MPS) introduced by Betcke and Trefethen [3] for Laplace eigenvalue problems on polygons, and has also been applied in the method of fundamental solutions [1]. We describe this approach in detail in Section 4.1.

1.2 Outline and contribution

This thesis is a numerical study of how the rectangular eigenvalue problem can be solved in practice, and of where the available techniques remain reliable. The remainder of this thesis is organized as follows.

Section 2 reviews the foundational mathematical tools used throughout the text, including matrix factorizations, perturbation bounds, and randomized sketching.

Section 3 establishes the core computational methodologies. Building on the minimum perturbation method for linear pencils, we assemble a complete approach for nonlinear rectangular problems that combines rational approximation, linearization of finite representations, and minimum perturbation extraction. As part of this approach we also introduce a cheap eigenvector recovery step that reuses the block structure of the linearized pencil to avoid a full singular value decomposition at each candidate.

Section 4 applies these tools to three distinct problems. Section 4.1 tests the scan-refinement approach on a Laplace eigenvalue problem on polygons, discretized by the method of particular solutions [3]. We show that this procedure is reliable once a cluster-aware post-processing step is added to separate closely spaced eigenvalues. On the other hand, the rational-approximation route struggles on this problem. Section 4.2 isolates the cause of this difficulty on a natively quadratic rectangular problem arising from a damped wave equation: by bypassing rational approximation and linearizing directly, we recover the physical eigenvalues accurately, suggesting that the extraction machinery itself works reliably when the companion pencil is of moderate size. Section 4.3 evaluates whether randomized sketching can reduce the cost of the minimum perturbation method, finding that it cannot achieve the precision of the unsketched baseline. A subsequent perturbation analysis explains this performance by showing how sketching scrambles the trailing singular subspace required for accurate extraction.

Contributions. We assemble and test a complete computational approach for nonlinear rectangular eigenvalue problems, combining rational approximation, companion linearization, and minimum perturbation extraction, and report the numerical difficulties encountered at each stage. We introduce a cheap diagnostic step that exploits the block structure of the linearized companion pencil to screen candidate eigenpairs before committing to a full singular value decomposition on the original problem. We develop a cluster-aware post-processing step for the scan-refinement method that resolves closely spaced eigenvalues without refining the global grid. Finally, we use multiplicative perturbation bounds together with numerical experiments to explain the accuracy loss when randomized sketching is applied to the minimum perturbation extraction.

2 Preliminaries

2.1 Notation

Throughout this thesis, we adopt the following conventions:

- We use $\mathbb{R}$ and $\mathbb{C}$ to denote the fields of real and complex numbers, respectively. The space of $m \times n$ complex matrices is denoted by $\mathbb{C}^{m\times n}$. We assume $m \geq n$ in the rest of the thesis.

- Matrices are denoted by uppercase Latin letters (A, Q, V). The lowercase letters x, y, u, v, c, w are reserved for column vectors. All other lowercase Latin letters denote scalars or indices. Subscripts follow the parent symbol: The i-th entry of a vector v is denoted by v_i. The (i, j)-th entry of a matrix A is denoted by A_{ij}, and ω_j is the j-th member of a scalar family. Greek letters are preferred for eigenvalues, singular values, and continuous parameters (λ, σ, ε). Matrix-valued functions carry their argument explicitly, e.g., $A(\lambda)$.

- For a complex scalar $z \in \mathbb{C}$, its complex conjugate is denoted by $\bar{z}$, and its real and imaginary parts are denoted by $\Re(z)$ and $\Im(z)$, respectively.
- The superscripts $(\cdot)^\top$ and $(\cdot)^\mathsf{H}$ denote the transpose and conjugate transpose of a matrix or vector, respectively.
- $\|\cdot\|_2$ denotes the 2-norm (spectral norm for matrices, Euclidean norm for vectors), and $\|\cdot\|_F$ denotes the Frobenius norm for matrices.
- $\mathrm{vec}(A)$ stacks the columns of A into a single vector, and $\otimes$ denotes the Kronecker product.
- $\mathcal{N}(0,1)$ denotes the standard normal distribution. A random matrix is said to have i.i.d. $\mathcal{N}(0,1)$ entries when each entry is drawn independently from this distribution.
- $\measuredangle(x, W)$ denotes the angle between a nonzero vector x and a subspace W, defined as (1).

2.2 Matrix factorizations

In this section, we briefly review two fundamental matrix factorizations that will be extensively used in the theoretical analysis and numerical implementation.

2.2.1 Singular value decomposition (SVD)

The singular value decomposition (SVD) is a ubiquitous tool for analyzing the properties of rectangular matrices. For any matrix $A \in \mathbb{C}^{m\times n}$ with $m \geq n$, its thin SVD is defined as

$$A = U\Sigma V^\mathsf{H},$$

where $U \in \mathbb{C}^{m\times n}$ has orthonormal columns, $V \in \mathbb{C}^{n\times n}$ is unitary, and $\Sigma \in \mathbb{R}^{n\times n}$ is a diagonal matrix containing the singular values of A. The singular values are real, non-negative, and conventionally ordered in descending magnitude:

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0.$$

The computational complexity for this thin SVD is $O(mn^2)$. In the context of rectangular eigenvalue problems, we are particularly interested in the smallest singular value, denoted by $\sigma_{\min}(A) := \sigma_n$.

Lemma 2.1 (Subspace angle bound)**.** *For a nonzero vector y and a subspace W, we define the acute angle between y and W by*

$$\measuredangle(y, W) = \arccos \frac{\|P_W y\|_2}{\|y\|_2} \in [0, \pi/2], \tag{1}$$

where P_W is the orthogonal projector onto W. Let $A \in \mathbb{C}^{m\times n}$ and $A = U\Sigma V^{\mathsf{H}}$ be its SVD decomposition described above with $\sigma_k > 0$. Let its right singular vectors be partitioned as $V = [V_k, V_\perp]$, where $V_k \in \mathbb{C}^{n\times k}$. Define $W = \mathrm{range}(V_\perp)$. Then, for any non-zero vector $y \in \mathbb{C}^n$,

$$\sin \measuredangle(y, W) \leq \frac{\|Ay\|_2}{\sigma_k \|y\|_2}.$$

Proof. Using $A^{\mathsf{H}}A = V\Sigma^2 V^{\mathsf{H}}$, we have

$$\begin{aligned} \|Ay\|_2^2 &\geq y^{\mathsf{H}}(V_k \Sigma_k^2 V_k^{\mathsf{H}})y \geq \sigma_k^2 y^{\mathsf{H}}(I - V_\perp V_\perp^{\mathsf{H}})y \\ &= \sigma_k^2(\|y\|_2^2 - \|V_\perp^{\mathsf{H}} y\|_2^2) = \sigma_k^2 \|y\|_2^2 \sin^2 \measuredangle(y, W). \end{aligned} \tag{2}$$

For $\sigma_k > 0$, dividing by $\sigma_k^2\|y\|_2^2$ and taking the square root completes the proof. □

2.2.2 Column-pivoted QR factorization (CPQR)

The QR factorization offers a more efficient alternative for orthogonalization. A standard thin QR factorization decomposes $A \in \mathbb{C}^{m\times n}$ into $A = QR$, where $Q \in \mathbb{C}^{m\times n}$ has orthonormal columns and $R \in \mathbb{C}^{n\times n}$ is upper triangular. However, the standard QR factorization is not reliably rank-revealing. When A is rank-deficient or ill-conditioned, we employ the column-pivoted QR factorization (CPQR). The CPQR introduces a permutation matrix $P \in \mathbb{R}^{n\times n}$ such that

$$AP = QR.$$

The permutation matrix P is typically chosen to ensure that the diagonal elements of the upper triangular matrix R are non-increasing in absolute value

$$|r_{11}| \geq |r_{22}| \geq \cdots \geq |r_{nn}| \geq 0.$$

The computational complexity for this CPQR is $O(mn^2)$. By greedily selecting columns with the largest remaining norms, the algorithm permutes nearly linearly dependent columns to the right. If A has numerical rank $k < n$, the trailing $n - k$ diagonal entries of R become negligible, revealing the rank.

2.3 Uniqueness and smoothness of the QR factorization

The standard thin QR factorization is not unique. For any diagonal unitary matrix D, the pair $\tilde{Q} = QD$, $\tilde{R} = D^{-1}R$ is equally valid. When A is rank-deficient, the ambiguity is more severe, because the columns of Q spanning the orthogonal complement of range(A) are entirely undetermined. Standard implementations fill this null-space component with arbitrary orthonormal vectors, so the factors of a continuously varying family $A(\lambda)$ can jump erratically.

Controlling these degrees of freedom is crucial for constructing a smooth rational surrogate from sampled QR bases, as required in Section 4.1 when applying `sketchAAA` to the boundary basis family $Q_B(\lambda)$.

Theorem 2.1 (Uniqueness of Normalized QR Factorization [18]). *Suppose $A \in \mathbb{C}^{m\times n}$ $(m \geq n)$ has full column rank. The thin QR factorization $A = QR$ is unique, where $Q \in \mathbb{C}^{m\times n}$ satisfying $Q^{\mathsf{H}}Q = I$ and $R \in \mathbb{C}^{n\times n}$ is upper triangular with strictly positive real diagonal entries.*

Proof. Suppose $A = Q_1R_1 = Q_2R_2$ are two normalized QR factorizations. Then $R_1^{\mathsf{H}}R_1 = A^{\mathsf{H}}A = R_2^{\mathsf{H}}R_2$, so $X = R_1R_2^{-1}$ satisfies $X = (X^{-1})^{\mathsf{H}}$. Since X is a product of upper triangular matrices it is upper triangular, while $(X^{-1})^{\mathsf{H}}$ is lower triangular. Hence $X = D$ is diagonal with positive real entries. The relation $D = D^{-1}$ then forces $D = I$, giving $R_1 = R_2$ and $Q_1 = AR_1^{-1} = AR_2^{-1} = Q_2$. □

For a smooth matrix family of constant full column rank, the smooth dependence of its unique QR factorization is guaranteed by the following theorem.

Theorem 2.2 ([13], Proposition 2.3). *Let $A(\lambda)$ be an $m \times n$ $(m \geq n)$ matrix-valued function of class C^k with full column rank for all λ. Then, it admits a QR factorization $A(\lambda) = Q(\lambda)R(\lambda)$ in which $Q(\lambda)$ is also of class C^k.*

2.4 The orthogonal Procrustes problem

While the normalized QR factorization guarantees the smoothness of an orthonormal basis $Q(\lambda)$, numerical algorithms often focus on discrete sample points λ_k. The strict fixed-sign constraint in normalized QR may induce unnecessarily large rotations within the column space, leading to suboptimal variation between consecutive discrete bases Q_k and Q_{k-1}. To minimize this point-to-point variation and extract a basis that varies as minimally as the underlying subspace itself, we aim to find a unitary transformation that optimally aligns a given basis to a reference basis without changing the column space it spans. This alignment process can be formulated as orthogonal Procrustes problem [18, §6.4.1].

Definition 2.1 (The Orthogonal Procrustes Problem). Given two matrices $Q_0, Q \in \mathbb{C}^{m\times n}$, the orthogonal Procrustes problem seeks a unitary matrix $W \in \mathbb{C}^{n\times n}$ that minimizes $\|QW - Q_0\|_F$.

Proposition 2.1. *The solution to Definition 2.1 is given by $W = VU^{\mathsf{H}}$, where U, V are from the SVD of $Q_0^{\mathsf{H}}Q = U\Sigma V^{\mathsf{H}}$.*

Proof. Using the definition of the Frobenius norm, we can write

$$\begin{aligned}
\operatorname*{argmin}_{W^{\mathsf{H}}W=I} \|QW - Q_0\|_F^2 &= \operatorname*{argmin}_{W^{\mathsf{H}}W=I} \operatorname{Tr}((QW - Q_0)(QW - Q_0)^{\mathsf{H}}) \\
&= \operatorname*{argmin}_{W^{\mathsf{H}}W=I} \operatorname{Tr}(QQ^{\mathsf{H}} - Q_0W^{\mathsf{H}}Q^{\mathsf{H}} - QWQ_0^{\mathsf{H}} + Q_0Q_0^{\mathsf{H}}) \\
&= \operatorname*{argmax}_{W^{\mathsf{H}}W=I} \left(\overline{\operatorname{Tr}(QWQ_0^{\mathsf{H}})} + \operatorname{Tr}(QWQ_0^{\mathsf{H}})\right) \\
&= \operatorname*{argmax}_{W^{\mathsf{H}}W=I} 2\Re \operatorname{Tr}(WQ_0^{\mathsf{H}}Q). \qquad (3)
\end{aligned}$$

Substituting the SVD $Q_0^{\mathsf{H}}Q = U\Sigma V^{\mathsf{H}}$, we can rewrite

$$\Re \operatorname{Tr}(WQ_0^{\mathsf{H}}Q) = \Re \operatorname{Tr}(WU\Sigma V^{\mathsf{H}}) = \Re \operatorname{Tr}(V^{\mathsf{H}}WU\Sigma).$$

Define $M = V^{\mathsf{H}}WU$, then M is a unitary matrix and $|M_{jj}| \le 1$ for $j = 1, 2, \ldots, n$. Consequently, its real part satisfies $\Re(M_{jj}) \le 1$, which gives

$$\Re\,\mathrm{Tr}(M\Sigma) = \sum_{j=1}^{n} \Re(M_{jj})\sigma_j \le \sum_{j=1}^{n} \sigma_j.$$

Therefore, the maximum is attained at $W = VU^{\mathsf{H}}$, yielding the solution to (3). □

Remark. Following [26, Theorem 2.9], the Procrustes correction can also be used to obtain a locally smooth representative of a parameter-dependent subspace. Suppose that $Q(t) \in \mathbb{C}^{m\times n}$ has orthonormal columns and spans a subspace with orthogonal projector $P(t) = Q(t)Q(t)^{\mathsf{H}}$. Fix a reference basis $Q_0 = Q(t_0)$ and define

$$\bar{Q}(t) = Q(t)W(t), \qquad W(t) = \arg\min_{W^{\mathsf{H}}W=I} \|Q(t)W - Q_0\|_F.$$

If $Q_0^{\mathsf{H}}Q(t)$ remains nonsingular in a neighbourhood of t_0, then the aligned basis $\bar{Q}(t)$ is locally as smooth as the projector $P(t)$.

2.5 Randomized sketching

In this subsection we introduce the two sketching matrices used in this work and recall the notion of an oblivious subspace embedding.

- A scaled Gaussian sketch is given by $S = \frac{1}{\sqrt{\ell}}G \in \mathbb{R}^{\ell\times m}$, where G has i.i.d. $\mathcal{N}(0, 1)$ entries. For a dense matrix $A \in \mathbb{R}^{m\times k}$, forming SA costs $O(\ell mk)$.
- A subsampled randomized Hadamard transform (SRHT) is given by

$$S = \sqrt{\frac{m}{\ell}}RHD \in \mathbb{R}^{\ell\times m},$$

  where $D = \mathrm{diag}(\xi_1, \ldots, \xi_m)$ with i.i.d. Rademacher entries $\xi_i \in \{-1, 1\}$, $H \in \mathbb{R}^{m\times m}$ is the normalized Walsh–Hadamard matrix with $H^\top H = I_m$, and $R \in \mathbb{R}^{\ell\times m}$ selects ℓ rows uniformly at random without replacement. The product HD can be applied by the fast Walsh–Hadamard transform in $O(m \log m)$ per column, so forming SA costs $O(mk \log m)$.

Definition 2.2 (Subspace embedding)**.** Let $V \subset \mathbb{R}^m$ be a fixed k-dimensional subspace. A matrix $S \in \mathbb{R}^{\ell\times m}$ is a γ-subspace embedding for V if

$$(1 - \gamma)\|x\|_2^2 \le \|Sx\|_2^2 \le (1 + \gamma)\|x\|_2^2$$

for all $x \in V$. Equivalently, if $U \in \mathbb{R}^{m\times k}$ is an orthonormal basis for V, then

$$\|U^\top S^\top SU - I_k\|_2 \le \gamma.$$

A random matrix S is called an oblivious subspace embedding (OSE) [39] with parameters (k, γ, δ) if it is a γ-embedding for any fixed k-dimensional subspace with probability at least $1-\delta$.

The OSE property holds for Gaussian sketches when $\ell = O(\gamma^{-2}(k + \log(1/\delta)))$ and for SRHT when $\ell = O(\gamma^{-1}\log(k/\delta)(k + \log(n/\delta)))$. In practice, $\ell = 2k$ is a common choice.

A direct consequence of the embedding property is that sketching preserves singular values. If S is a γ-subspace embedding for range(A), then

$$\sqrt{1-\gamma}\,\sigma_i(A) \leq \sigma_i(SA) \leq \sqrt{1+\gamma}\,\sigma_i(A), \tag{4}$$

for $i = 1, \ldots, n$, as follows directly from substituting the embedding inequality into the Courant–Fischer min-max characterization of $\sigma_i(SA)^2$.

While the results above characterize the required sketch dimension, we need explicit upper bounds on the distortion ε with respect to the sketch dimension ℓ for the analysis in Section 4.3.

Gaussian. Let $U \in \mathbb{R}^{m\times k}$ be a matrix with orthonormal columns, and let $G \in \mathbb{R}^{\ell\times m}$ be a matrix with i.i.d. standard normal entries. Then GU is an $\ell \times k$ matrix with independent standard normal entries, and the inequality for singular values of $SU = \frac{1}{\sqrt{\ell}}GU$

$$1 - \sqrt{\frac{k}{\ell}} - \sqrt{\frac{2\ln(2/\delta)}{\ell}} \leq \sigma_{\min}(SU) \leq \sigma_{\max}(SU) \leq 1 + \sqrt{\frac{k}{\ell}} + \sqrt{\frac{2\ln(2/\delta)}{\ell}}.$$

holds with probability at least $1-\delta$ [35]. Setting $\alpha = \sqrt{k/\ell} + \sqrt{2\ln(2/\delta)/\ell}$, the distortion satisfies

$$\|U^\top(S^\top S - I_m)U\|_2 = \max\{\sigma_{\max}^2(SU) - 1,\ 1 - \sigma_{\min}^2(SU)\} \leq \alpha(2+\alpha). \tag{5}$$

In particular, this implies that the distortion remains of order $\sqrt{k/m}$ even at $\ell = m$.

SRHT. To capture the behavior at ℓ close to m, we prove a bound that measures the contribution of the deleted rows.

Lemma 2.2. *Let $U \in \mathbb{R}^{m\times k}$ be a matrix with orthonormal columns, $S = \sqrt{\frac{m}{\ell}}RHD \in \mathbb{R}^{\ell\times m}$ an SRHT matrix, and $E = I_m - R^\top R$ the projector onto the $m-\ell$ deleted coordinates. Denote by $w_i = W_{i,:}^\top \in \mathbb{R}^k$ the transpose of i-th row of $W = HDU \in \mathbb{R}^{m\times k}$. If the row-norm property [37, Lemma 3.3]*

$$\max_{1\leq i\leq m} \|w_i\|_2^2 \leq \mu := \frac{1}{m}(\sqrt{k} + \sqrt{8\ln(m/\delta)})^2$$

holds with probability at least $1-\delta$, then

$$\|U^\top(S^\top S - I_m)U\|_2 \leq \frac{m-\ell}{\ell}\left(1 + (\sqrt{k} + \sqrt{8\ln(m/\delta)})^2\right). \tag{6}$$

Proof. Since HD is orthogonal, $W = HDU$ has orthonormal columns. Using $R^\top R + E = I_m$,

$$S^\top S - I_m = \frac{m}{\ell} DH^\top (I_m - E)HD - I_m = \frac{m-\ell}{\ell} I_m - \frac{m}{\ell} DH^\top EHD. \tag{7}$$

Let $\mathcal{E} \subset \{1, \dots, m\}$, $|\mathcal{E}| = m - \ell$ index the deleted rows. Since $E = \sum_{i\in\mathcal{E}} e_i e_i^\top$,

$$U^\top DH^\top EHD\, U = W^\top EW = \sum_{i\in\mathcal{E}} w_i w_i^\top.$$

Restricting (7) to the column space of U and applying the triangle inequality gives

$$\|U^\top (S^\top S - I_m)U\|_2 \le \frac{m-\ell}{\ell} + \frac{m}{\ell} \|\sum_{i\in\mathcal{E}} w_i w_i^\top\|_2 \le \frac{m-\ell}{\ell} + \frac{m}{\ell} \sum_{i\in\mathcal{E}} \|w_i\|_2^2 \le \frac{m-\ell}{\ell} + \frac{m}{\ell}(m-\ell)\mu.$$

Substituting $\mu = \frac{1}{m}(\sqrt{k} + \sqrt{8\ln(m/\delta)})^2$ gives (6). □

For fixed δ and m, the bound (6) decreases as the number of retained rows ℓ increases, and vanishes at $\ell = m$.

2.6 Matrix perturbation theory

In this subsection, we introduce fundamental bounds that quantify how eigenvalues and their corresponding invariant subspaces shift under additive and multiplicative matrix perturbations.

Theorem 2.3 (Weyl's Inequality for singular values [5, Theorem 5.41])**.** *Let $A, H \in \mathbb{C}^{m\times n}$. For any $k = 1, 2, \dots, \min\{m, n\}$, the absolute difference between their k-th singular values is bounded by*

$$|\sigma_k(A + H) - \sigma_k(A)| \le \|H\|_2.$$

While Weyl's inequalities bound the perturbation of individual eigenvalues and singular values, one sometimes needs to control the rotation of the corresponding invariant subspaces. We first recall the standard notion of principal angles, and then present a multiplicative perturbation bound for singular subspaces under sketching. The latter result is essentially from Park and Nakatsukasa [34], building on the relative perturbation theory of Li [29]. We restate it here in the notation of this thesis and derive the corollary that will be applied in Section 4.3.

Definition 2.3 (Principal angles)**.** Let $X, Y \in \mathbb{R}^{m\times n}$ have orthonormal columns. The principal angles $0 \le \theta_1 \le \cdots \le \theta_n \le \pi/2$ between range(X) and range(Y) are defined by $\cos\theta_i = \sigma_i(X^\top Y)$. We denote

$$\sin\Theta(X, Y) = \mathrm{diag}(\sin\theta_1, \dots, \sin\theta_n).$$

If $X_\perp$ is an orthonormal basis of the orthogonal complement of range(X), then

$$\|\sin\Theta(X, Y)\|_2 = \|X_\perp^\top Y\|_2.$$

When a matrix is left-multiplied by a sketching matrix, the resulting perturbation is multiplicative rather than additive, thus the classical Davis–Kahan theorem [11] can be unnecessarily pessimistic. The following results provide sharper bounds in this setting.

Definition 2.4 (Relative gap)**.** For $a, b \in \mathbb{R}$, define

$$\chi(a, b) = \frac{|a-b|}{\sqrt{|ab|}},$$

with the conventions $0/0 = 0$ and $1/0 = \infty$.

The following theorem bounds the rotation of the trailing right singular subspace when a matrix is left-multiplied by another matrix.

Theorem 2.4 ([34, Theorem 3.1])**.** *Let $A \in \mathbb{R}^{m\times n}$ with $m \geq n$ have thin SVD $A = U\Sigma V^\top$, where $U \in \mathbb{R}^{m\times n}$, $\Sigma \in \mathbb{R}^{n\times n}$, $V \in \mathbb{R}^{n\times n}$, with singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Let $S \in \mathbb{R}^{\ell\times m}$ with $\ell \geq n$ be such that SU has full column rank, and let $SU = QR$ be a thin QR factorization with $Q \in \mathbb{R}^{\ell\times n}$, $R \in \mathbb{R}^{n\times n}$. Denote the singular values of $\tilde{A} = SA$ by $\tilde{\sigma}_1 \geq \cdots \geq \tilde{\sigma}_n \geq 0$, and partition the right singular vectors of A and $\tilde{A}$ as*

$$V = [V_1, V_2], \qquad \tilde{V} = [\tilde{V}_1, \tilde{V}_2],$$

where $V_1, \tilde{V}_1 \in \mathbb{R}^{n\times(n-k)}$ correspond to the $(n-k)$ largest singular values and $V_2, \tilde{V}_2 \in \mathbb{R}^{n\times k}$ to the k smallest. If there exist $\alpha > 0$ and $\delta > 0$ such that

$$\min_{1\leq i\leq n-k} \sigma_i \geq \alpha + \delta \qquad \text{and} \qquad \max_{1\leq j\leq k} \tilde{\sigma}_{n-k+j} \leq \alpha,$$

or

$$\min_{1\leq i\leq n-k} \tilde{\sigma}_i \geq \alpha + \delta \qquad \text{and} \qquad \max_{1\leq j\leq k} \sigma_{n-k+j} \leq \alpha,$$

then

$$\|\sin\Theta(V_2, \tilde{V}_2)\|_2 \leq \frac{\|R - R^{-\top}\|_2}{\chi(\alpha^2,\ (\alpha+\delta)^2)}.$$

The numerator $\|R - R^{-\top}\|_2$ measures how far the sketching matrix deforms the left singular basis from orthonormality. The following lemma connects it to the subspace-embedding distortion γ defined in Section 2.5.

Lemma 2.3. *Let $U \in \mathbb{R}^{m\times n}$ have orthonormal columns and let $S \in \mathbb{R}^{\ell\times m}$ be a γ-subspace embedding for* $\operatorname{range}(U)$ *with $\gamma < 1$. Let $SU = QR$ be a thin QR factorization. Then*

$$\|R - R^{-\top}\|_2 \leq \frac{\gamma}{\sqrt{1-\gamma}}.$$

Proof. Since $R^\top R = U^\top S^\top S\,U$, the embedding property (4) gives $\sigma_i(R)^2 \in [1-\gamma, 1+\gamma]$, so in particular R is invertible. Writing the SVD $R = W\Sigma_R Z^\top$, we have $R^{-\top} = W\Sigma_R^{-1}Z^\top$, so $\|R - R^{-\top}\|_2 = \max_i |\sigma_i(R) - \sigma_i(R)^{-1}|$. The function $t \mapsto |t - t^{-1}|$ is decreasing on $(0, 1]$, hence

$$\max_i |\sigma_i(R) - \sigma_i(R)^{-1}| \leq \frac{1}{\sqrt{1-\gamma}} - \sqrt{1-\gamma} = \frac{\gamma}{\sqrt{1-\gamma}}. \qquad \square$$

Combining Theorem 2.4 and Lemma 2.3 yields an a priori bound in terms of the original singular values and the embedding distortion alone.

Corollary 2.1. *In the setting of Theorem 2.4, suppose S is a γ-subspace embedding for* range(A) *with* $(1-\gamma)\,\sigma_{n-k}^2 > \sigma_{n-k+1}^2$. *Then*

$$\|\sin\Theta(\tilde{V}_2, V_2)\|_2 \le \frac{\gamma\,\sigma_{n-k}\,\sigma_{n-k+1}}{(1-\gamma)\,\sigma_{n-k}^2 - \sigma_{n-k+1}^2}. \tag{8}$$

Proof. We apply Theorem 2.4 using its second set of hypotheses. Set $\alpha = \sigma_{n-k+1}$ and $\alpha + \delta = \sqrt{1-\gamma}\,\sigma_{n-k}$. The condition $\max_{1\le j\le k}\sigma_{n-k+j} \le \alpha$ holds trivially since $\sigma_{n-k+1} \ge \sigma_{n-k+2} \ge \cdots \ge \sigma_n$. By (4), $\tilde{\sigma}_{n-k} \ge \sqrt{1-\gamma}\,\sigma_{n-k} = \alpha+\delta$, so the condition $\min_{1\le i\le n-k}\tilde{\sigma}_i \ge \alpha+\delta$ is satisfied. The relative gap evaluates to

$$\chi(\alpha^2, (\alpha+\delta)^2) = \frac{(1-\gamma)\sigma_{n-k}^2 - \sigma_{n-k+1}^2}{\sqrt{1-\gamma}\,\sigma_{n-k}\sigma_{n-k+1}},$$

and dividing the numerator bound $\|R - R^{-\top}\|_2 \le \gamma/\sqrt{1-\gamma}$ gives the result. □

With these foundational tools in place, we now turn to the rectangular eigenvalue problem itself and introduce the computational methods that form the core of this thesis.

3 Rectangular Eigenvalue Problem

In this section, we introduce the core computational machinery for solving rectangular eigenvalue problems. We begin in Section 3.1 with the minimum perturbation method for linear pencils, which reduces the rectangular problem to a standard generalized eigenvalue problem via a total least-squares formulation. Section 3.2 then describes how rational approximation can compress a nonlinear matrix-valued function into a finite surrogate, and Section 3.3 shows how such surrogates as well as other natively finite representations can be linearized into rectangular companion pencils amenable to the minimum perturbation method.

3.1 Minimum perturbation method

For a generalized rectangular linear pencil $M - \lambda B$, where $M, B \in \mathbb{C}^{m\times n}$ with $m > n$, the pencil may fail to have n linearly independent eigenvectors. We therefore seek the nearest perturbed pencil $\hat{M} - \lambda\hat{B}$ that admits n linearly independent eigenvectors [8, 25]. This gives

$$\begin{cases} \text{minimize} & \|\hat{M} - M\|_F^2 + \|\hat{B} - B\|_F^2, \\ \text{subject to} & \hat{M}x_k = \lambda_k\hat{B}x_k, \quad k = 1,\dots,n, \\ & x_1,\dots,x_n \text{ linearly independent.} \end{cases}$$

Let $X = [x_1,\dots,x_n]$ and $\Lambda = \mathrm{diag}(\lambda_1,\dots,\lambda_n)$. Since X is nonsingular, the eigenvalue constraint is equivalent to $\hat{M} = \hat{B}Z$, where $Z = X\Lambda X^{-1}$. Conversely, if Z is diagonalizable, say $Z = X\Lambda X^{-1}$ with X nonsingular, then $\hat{M} = \hat{B}Z$ implies $\hat{M}X = \hat{B}X\Lambda$, so the columns of X are n linearly independent eigenvectors of the perturbed pencil. Hence the minimum perturbation problem can be written as

$$\begin{cases} \text{minimize} & \|[\hat{B}, \hat{M}] - [B, M]\|_F^2, \\ \text{subject to} & \hat{M} = \hat{B}Z, \quad Z \text{ diagonalizable.} \end{cases} \tag{9}$$

Dropping the diagonalizability constraint gives the multidimensional total least squares (TLS) problem

$$\begin{cases} \text{minimize} & \|[\hat{B}, \hat{M}] - [B, M]\|_F^2, \\ \text{subject to} & \hat{M} = \hat{B}Z. \end{cases} \tag{10}$$

We define the partition for the singular value decomposition of $[B, M]$ as

$$[B, M] = [U_1\ U_2] \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix} \begin{bmatrix} V_1^{\mathrm{H}} \\ V_2^{\mathrm{H}} \end{bmatrix} = [U_1\ U_2] \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix} \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}^{\mathrm{H}}, \tag{11}$$

where $U_1 \in \mathbb{C}^{m\times n}$, $\Sigma_1 \in \mathbb{R}^{n\times n}$, and $V_{ij} \in \mathbb{C}^{n\times n}$. Then the following theorem gives the solution of problem (10).

Theorem 3.1 ([38], Theorem 3.1)**.** *If $\sigma_n([B, M]) > \sigma_{n+1}([B, M])$ and V_{22} is nonsingular, then the multidimensional TLS problem (10) has the unique solution*

$$[\Delta B_0, \Delta M_0] = -\,U_2\Sigma_2[V_{12}^{\mathrm{H}}, V_{22}^{\mathrm{H}}], \tag{12}$$

where $\Delta B_0 = \hat{B} - B$, $\Delta M_0 = \hat{M} - M$. The corresponding rank-n corrected matrix is $[\hat{B}, \hat{M}] = U_1\Sigma_1[V_{11}^{\mathrm{H}}, V_{21}^{\mathrm{H}}]$ and $Z_0 = -V_{12}V_{22}^{-1}$ gives the matrix Z in (10).

Proof. Since $\hat{M} = \hat{B}Z$, we have $[\hat{B}, \hat{M}] = \hat{B}[I, Z]$, and hence $\operatorname{rank}([\hat{B}, \hat{M}]) \le n$. Let $[B, M] = U\Sigma V^{\mathrm{H}}$ be the singular value decomposition partitioned as (11). By the Eckart–Young–Mirsky theorem,

$$\|[B, M] - [\hat{B}, \hat{M}]\|_F = \|[\Delta B, \Delta M]\|_F \ge \sqrt{\sum_{i=n+1}^{d} \sigma_i^2},$$

where $d = \operatorname{rank}([B, M])$, and the equality holds when $[\Delta B, \Delta M]$ is given by (12). The condition $\sigma_n([B, M]) > \sigma_{n+1}([B, M])$ gives uniqueness of this rank-n approximation $[\hat{B}, \hat{M}]$.

It remains to recover Z. Since $[\hat{B}, \hat{M}]$ has rank n, its right null space is spanned by $\begin{bmatrix} V_{12} \\ V_{22} \end{bmatrix}$. On the other hand, $\hat{M} = \hat{B}Z$ is equivalent to $[\hat{B}, \hat{M}] \begin{bmatrix} Z \\ -I \end{bmatrix} = 0$. Hence there exists a nonsingular matrix $K \in \mathbb{C}^{n\times n}$ such that

$$\begin{bmatrix} Z \\ -I \end{bmatrix} = \begin{bmatrix} V_{12} \\ V_{22} \end{bmatrix} K.$$

From the lower block we get $-I = V_{22}K$. Since V_{22} is nonsingular, we have $K = -V_{22}^{-1}$ and

$Z_0 = -V_{12}V_{22}^{-1}$. □

We now build the relation between the TLS problem and the minimum perturbation problem.

Theorem 3.2 ([25], Theorem 2)**.** *The minimum perturbation problem* (9) *and the TLS problem* (10) *have the same infimum value. Moreover, under the assumptions of Theorem 3.1, if the TLS minimizer $Z_0 = -V_{12}V_{22}^{-1}$ is diagonalizable, then the perturbations $\Delta B_0, \Delta M_0$ attain the minimum perturbation problem* (9)*.*

Proof. We compare the feasible sets at the level of perturbed block matrices $[\hat{B}, \hat{M}]$. Let $\mathcal{P}_{\mathrm{mp}}$ be the set of matrices $[\hat{B}, \hat{M}]$ such that $\hat{M} = \hat{B}Z$ for some diagonalizable $Z \in \mathbb{C}^{n\times n}$, and let $\mathcal{P}_{\mathrm{tls}}$ be the corresponding set where Z is arbitrary. Clearly, $\mathcal{P}_{\mathrm{mp}} \subseteq \mathcal{P}_{\mathrm{tls}}$.

We show that $\mathcal{P}_{\mathrm{tls}} \subseteq \overline{\mathcal{P}_{\mathrm{mp}}}$. Take any $[\hat{B}, \hat{M}] \in \mathcal{P}_{\mathrm{tls}}$. Then $[\hat{B}, \hat{M}] = \hat{B}[I, Z]$ for some $Z \in \mathbb{C}^{n\times n}$. Let $Z = QTQ^{\mathsf{H}}$ be a Schur decomposition, where Q is unitary and T is upper triangular. For any $\varepsilon > 0$, choose $D_\varepsilon = \mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_n)$ such that the diagonal entries of $T_\varepsilon = T + D_\varepsilon$ are pairwise distinct and

$$\varepsilon_i < \frac{\varepsilon}{1 + \|\hat{B}\|_F}.$$

Then T_ε has distinct eigenvalues and is therefore diagonalizable. Hence $Z_\varepsilon = QT_\varepsilon Q^{\mathsf{H}}$ is diagonalizable and

$$\|[\hat{B}, \hat{M}] - [\hat{B}, \hat{B}Z_\varepsilon]\|_F = \|\hat{B}(Z - Z_\varepsilon)\|_F = \|\hat{B}QD_\varepsilon Q^{\mathsf{H}}\|_F \leq \|\hat{B}\|_F\|D_\varepsilon\|_2 < \varepsilon.$$

Thus $[\hat{B}, \hat{B}Z_\varepsilon] \in \mathcal{P}_{\mathrm{mp}}$ and it can be made arbitrarily close to $[\hat{B}, \hat{M}]$. Therefore

$$\mathcal{P}_{\mathrm{mp}} \subseteq \mathcal{P}_{\mathrm{tls}} \subseteq \overline{\mathcal{P}_{\mathrm{mp}}}.$$

Since the objective $\|[\hat{B}, \hat{M}] - [B, M]\|_F^2$ is continuous, the two problems have the same infimum value.

Under the assumptions of Theorem 3.1, the TLS problem has the unique minimizer $(\Delta B_0, \Delta M_0, Z_0)$. If Z_0 is diagonalizable, then this TLS minimizer satisfies the diagonalizability constraint in (9). Since the two problems have the same infimum value, it also attains the minimum perturbation problem. □

For computation, it is not necessary to form $Z_0 = -V_{12}V_{22}^{-1}$. From Theorem 3.1, the corrected block matrix is $[\hat{B}, \hat{M}] = U_1\Sigma_1[V_{11}^{\mathsf{H}}, V_{21}^{\mathsf{H}}]$. Thus $(\hat{M} - \lambda\hat{B})x = 0$ is equivalent to

$$U_1\Sigma_1[V_{11}^{\mathsf{H}}, V_{21}^{\mathsf{H}}]\begin{bmatrix} -\lambda x \\ x \end{bmatrix} = 0.$$

Since $U_1\Sigma_1$ has full column rank, this reduces to the generalized eigenvalue problem $V_{21}^{\mathsf{H}}x = \lambda V_{11}^{\mathsf{H}}x$. The procedure of the minimum perturbation method is summarized in Algorithm 1. The dominant cost is the thin SVD of $[B, M] \in \mathbb{C}^{m\times 2n}$, which is $O(mn^2)$.

Algorithm 1 Minimum perturbation method via TLS

1: **Input:** $M, B \in \mathbb{C}^{m\times n}$ from rectangular pencil $M - \lambda B$
2: Compute the singular value decomposition of $[B, M]$, denoted by $[B, M] = U\Sigma V^{\mathsf{H}}$
3: Solve $V_{21}^{\mathsf{H}} x = \lambda V_{11}^{\mathsf{H}} x$ and output the eigenpairs ▷ V_{11} and V_{21} are defined as in (11)

Discussion. The preceding construction shows that the perturbed correction produces a rank-n pencil $\hat{M} - \lambda\hat{B}$ whose eigenpairs are obtained from the square pencil $V_{21}^{\mathsf{H}} - \lambda V_{11}^{\mathsf{H}}$. We now relate these computed eigenpairs to residuals of the original rectangular pencil.

Suppose first that the original pencil admits an approximate eigenpair (λ, x) with $\|x\|_2 = 1$, such that $\|(M - \lambda B)x\|_2 = \varepsilon$. Let $C = [B, M]$, $y = \begin{bmatrix} -\lambda x \\ x \end{bmatrix}$, we have

$$\sigma_{\min}(C) \le \frac{\|Cy\|_2}{\|y\|_2} = \frac{\|(M - \lambda B)x\|_2}{\|y\|_2} = \frac{\varepsilon}{\sqrt{1 + |\lambda|^2}}.$$

Thus a small residual pair gives at least one small singular direction of the augmented matrix C. Define $\hat{C} = [\hat{B}, \hat{M}] = U_1\Sigma_1 \begin{bmatrix} V_{11}^{\mathsf{H}} & V_{21}^{\mathsf{H}} \end{bmatrix}$ as the rank-n corrected matrix. For the same vector y, we have

$$\hat{C}y = U_1\Sigma_1(V_{21}^{\mathsf{H}} - \lambda V_{11}^{\mathsf{H}})x.$$

Since $C = \hat{C} + U_2\Sigma_2[V_{12}^{\mathsf{H}}, V_{22}^{\mathsf{H}}]$ is the SVD decomposition, the two terms are orthogonal in the left singular basis. Hence

$$\|\Sigma_1(V_{21}^{\mathsf{H}} - \lambda V_{11}^{\mathsf{H}})x\|_2 \le \|\hat{C}y\|_2 \le \|Cy\|_2 = \varepsilon.$$

It follows that when $\sigma_n(C) > 0$,

$$\|(V_{21}^{\mathsf{H}} - \lambda V_{11}^{\mathsf{H}})x\|_2 \le \frac{\varepsilon}{\sigma_n(C)}.$$

Thus an approximate eigenpair of the original rectangular pencil also gives an approximate eigenpair of the reduced generalized eigenvalue problem. Equivalently, the augmented direction y lies within angle $\varepsilon/(\sigma_n(C)\sqrt{1 + |\lambda|^2})$ of the trailing right singular subspace span(V_2) following (2).

Conversely, let $(\hat{\lambda}, \hat{x})$ be an eigenpair of $Z_0 = -V_{12}V_{22}^{-1}$. Define $\hat{z} = V_{22}^{-1}\hat{x}$, then

$$V_2\hat{z} = \begin{bmatrix} V_{12} \\ V_{22} \end{bmatrix}\hat{z} = \begin{bmatrix} -\hat{\lambda}\hat{x} \\ \hat{x} \end{bmatrix}.$$

Therefore

$$(M - \hat{\lambda}B)\hat{x} = C\begin{bmatrix} -\hat{\lambda}\hat{x} \\ \hat{x} \end{bmatrix} = CV_2\hat{z} = U_2\Sigma_2\hat{z}.$$

Since V_2 has orthonormal columns,

$$\|\hat{z}\|_2 = \|V_2\hat{z}\|_2 = \sqrt{1 + |\hat{\lambda}|^2}\,\|\hat{x}\|_2.$$

Thus

$$\|(M - \hat{\lambda}B)\hat{x}\|_2 = \|\Sigma_2\hat{z}\|_2 \le \sigma_{n+1}(C)\sqrt{1 + |\hat{\lambda}|^2}\|\hat{x}\|_2.$$

This worst-case bound highlights a structural limitation of the TLS extraction. Even if the original pencil admits a highly accurate approximate eigenpair, the reduced GEP might fail to recover it. We now give a simple example illustrating this behavior, where the original pencil has a good approximate eigenpair, but the nonnormality of Z_0 drives the residuals of all eigenpairs extracted from the minimum perturbation method to the maximum bound.

Example 3.1. *Let* $0 < \varepsilon \ll 1$, $\gamma = \sqrt{1+\varepsilon^2}$, *and define*

$$C = [B, M] = \frac{1}{\gamma}\begin{bmatrix} 2 & 0 & 0 & 2\varepsilon \\ 0 & 2\varepsilon & 2 & 0 \\ 0 & -1 & \varepsilon & 0 \\ -\varepsilon^2 & 0 & 0 & \varepsilon \end{bmatrix}.$$

We first note that the original pencil has at least one small-residual direction. Indeed, for $x = e_2$ *and* $\lambda = 0$,

$$(M - \lambda B)x = Me_2 = \frac{1}{\gamma}\begin{bmatrix} 2\varepsilon \\ 0 \\ 0 \\ \varepsilon \end{bmatrix},$$

thus $\|(M - \lambda B)x\|_2 = \sqrt{5}\varepsilon/\gamma = O(\varepsilon)$.

We now apply the minimum perturbation extraction. Since the rows of C are mutually orthogonal with norms $2, 2, 1, \varepsilon$, *the right singular vectors corresponding to the two smallest singular values are*

$$V_2 = \frac{1}{\gamma}\begin{bmatrix} 0 & -\varepsilon \\ -1 & 0 \\ \varepsilon & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} V_{12} \\ V_{22} \end{bmatrix}.$$

The TLS construction gives

$$Z_0 = -V_{12}V_{22}^{-1} = \begin{bmatrix} 0 & \varepsilon \\ \varepsilon^{-1} & 0 \end{bmatrix}.$$

Therefore the computed eigenvalues are $\hat{\lambda}_\nu = \nu$, $\nu = \pm 1$, *with corresponding eigenvectors* $\hat{x}_\nu = (\varepsilon, \nu)^\top$. *However, these computed pairs have large residuals in the original pencil. Direct substitution gives*

$$(M - \hat{\lambda}_\nu B)\hat{x}_\nu = \begin{bmatrix} 0 \\ 0 \\ \gamma \\ \nu\varepsilon\gamma \end{bmatrix}.$$

Since $\|\hat{x}_v\|_2 = \gamma$, the normalized residual is

$$\frac{\|(M - \hat{\lambda}_v B)\hat{x}_v\|_2}{\|\hat{x}_v\|_2} = \sqrt{1+\varepsilon^2}.$$

Thus both computed eigenpairs have residual bounded below by 1, independently of ε.

Moreover, the original pencil does not have good approximate eigenpairs at the computed eigenvalues $\hat{\lambda} = \pm 1$. For any $x = (\alpha, \beta)^\top$,

$$\|(M - B)x\|_2^2 = \frac{1}{1+\varepsilon^2}\left(8|\alpha - \varepsilon\beta|^2 + (1+\varepsilon^2)|\varepsilon\alpha + \beta|^2\right) \geq (1+\varepsilon^2)\|x\|_2^2.$$

Therefore $\sigma_{\min}(M - B) \geq \sqrt{1+\varepsilon^2}$. Analogously, we have $\sigma_{\min}(M + B) \geq \sqrt{1+\varepsilon^2}$. The eigenvalues $\hat{\lambda} = \pm 1$ returned by the minimum perturbation extraction are therefore not approximate eigenvalues of the original rectangular pencil.

This example demonstrates that the minimum perturbation extraction produces eigenpairs of the corrected pencil, but these eigenpairs need not have small residuals in the original rectangular pencil. Therefore, it is essential in practice to monitor the residual to verify the reliability of the computed eigenpairs.

3.2 Rational approximation

We next describe how rational approximation can be used to construct a finite surrogate for a nonlinear rectangular matrix-valued function. Let $F(z) \in \mathbb{C}^{m\times n}$. The goal of this step is to approximate F by a rational function R that can be evaluated from a finite set of sampled matrices $F(z_j)$. In the next subsection, we show how such a finite representation, once written in a recurrence basis, can be converted into a rectangular linear pencil.

`AAA` rational approximation. The `AAA` [33] algorithm constructs rational approximants in barycentric form. For a scalar function f, given support points $z_0, \ldots, z_d$, function values $f(z_j)$, and weights ω_j, the barycentric approximant is

$$r(z) = \frac{\sum_{j=0}^{d} \dfrac{\omega_j f(z_j)}{z - z_j}}{\sum_{j=0}^{d} \dfrac{\omega_j}{z - z_j}}.$$

The support points are selected adaptively, and the weights are obtained from a least-squares problem involving a Loewner matrix.

`SketchAAA` for matrix-valued functions. For a matrix-valued function $F(z) \in \mathbb{C}^{m\times n}$, the same scalar weights and support points are used for all entries. Formally, one may apply AAA to the vectorized function $\boldsymbol{f}(z) = \mathrm{vec}(F(z)) \in \mathbb{C}^{mn}$. However, applying AAA directly to all mn components can be expensive when m and n are large. `SketchAAA` [19] reduces this cost by applying `AAA` to a low-dimensional random projection of $\boldsymbol{f}$.

Let $S \in \mathbb{C}^{mn\times\ell}$ be a random probing matrix with $\ell \ll mn$, and define the sketched function $\boldsymbol{g}(z) = S^*\boldsymbol{f}(z) \in \mathbb{C}^{\ell}$. `AAA` is then applied to $\boldsymbol{g}$, producing support points $z_0, \ldots, z_d$ and weights $\omega_0, \ldots, \omega_d$. These support points and weights are subsequently used to construct a matrix-valued barycentric approximant to F.

In this work, we use the tensorized probing strategy

$$S = [\boldsymbol{u}_1 \otimes \boldsymbol{v}_1, \cdots, \boldsymbol{u}_\ell \otimes \boldsymbol{v}_\ell],$$

where $\boldsymbol{u}_k \in \mathbb{C}^n$ and $\boldsymbol{v}_k \in \mathbb{C}^m$ are random vectors. The corresponding sketched components satisfy

$$g_k(z) = (\boldsymbol{u}_k \otimes \boldsymbol{v}_k)^* \operatorname{vec}(F(z)) = \boldsymbol{v}_k^* F(z)\boldsymbol{u}_k.$$

Thus the probing step can be implemented directly through bilinear forms on $F(z)$, without explicitly forming $\operatorname{vec}(F(z))$.

After the support points and weights have been computed from the sketched problem, we form the matrix-valued barycentric rational approximant

$$R_b(z) = \frac{\sum_{j=0}^{d} \dfrac{\omega_j F(z_j)}{z - z_j}}{\sum_{j=0}^{d} \dfrac{\omega_j}{z - z_j}}. \tag{13}$$

Rational Newton form. Starting from (13), the barycentric-to-Newton conversion [16, Theorem 3] gives

$$R_n(z) = \sum_{j=0}^{d} \eta_j(z) D_j, \tag{14}$$

where $D_j = F(z_j)$ and the rational Newton basis functions satisfy

$$\eta_0(z) \equiv 1, \qquad \eta_j(z) = \frac{z - \sigma_{j-1}}{\beta_j(h_j - k_j z)}\, \eta_{j-1}(z), \qquad j = 1, \ldots, d.$$

The Newton parameters are obtained explicitly from the barycentric data by

$$\sigma_{j-1} = z_{j-1}, \qquad \beta_j k_j = -\frac{\omega_{j-1}}{\omega_j}, \qquad \beta_j h_j = -\frac{z_j \omega_{j-1}}{\omega_j}, \qquad j = 1, \ldots, d.$$

3.3 Linearization of finite representations

We now describe how to convert certain finite representations of a matrix-valued function into a rectangular linear pencil. For square nonlinear eigenvalue problems, linearization is a standard way to replace a polynomial or rational finite representation by a generalized eigenvalue problem [21, 31]. While these linearization techniques were originally developed for square matrix-valued functions, the underlying algebraic transformations depend primarily on the scalar basis functions. Consequently, the mechanism can be extended naturally to the rectangular case.

To bridge the notation between rational approximation and the linear pencil, we now replace the complex domain variable z from Section 3.2 with the standard eigenvalue parameter λ. The approximation support points z_j remain fixed constants. Consider a matrix-valued function

$$G(\lambda) = \sum_{j=0}^{d} \alpha_j(\lambda) D_j,$$

where $D_j \in \mathbb{C}^{m\times n}$ are constant coefficient matrices and $\alpha_j(\lambda)$ are scalar basis functions. To design a linearization, we introduce auxiliary block variables representing the basis terms applied to the same vector x, for example $y^{(j)} = \alpha_j(\lambda)x$. The scalar recurrence relations satisfied by the functions α_j are then written as linear relations between these blocks. Combining these recurrence relations with the original equation $G(\lambda)x = 0$ gives a block pencil

$$(\mathcal{M} - \lambda\mathcal{B})y = 0.$$

This equivalence only depends on block structure and therefore remains valid when the coefficient matrices are rectangular. Conversely, if $G(\lambda)x$ is small and the auxiliary blocks are chosen consistently with the recurrence, then the residual of the linearized pencil is small as well, up to the scaling introduced by the chosen companion form. This follows from the same algebraic identity that underlies the equivalence in the square case [31].

We now detail this mechanism for three cases used in this work: a quadratic matrix-valued function, the rational barycentric representation and the rational Newton representation.

Quadratic matrix-valued functions. As a first example, consider a quadratic matrix-valued function

$$G(\lambda) = K + \lambda D + \lambda^2 M,$$

where the coefficient matrices are $K, D, M \in \mathbb{C}^{m\times n}$. To construct the rectangular linearization following the approach in [21], we seek a scalar parameter λ and a nonzero vector $x \in \mathbb{C}^n$ such that $G(\lambda)x = 0$. Introduce the enlarged vector

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \lambda x \\ x \end{bmatrix} \in \mathbb{C}^{2n}.$$

Using these variables, the original quadratic condition can be rewritten as the first-order relation $Dy_1 + Ky_2 + \lambda M y_1 = 0$. Coupling this with the defining auxiliary relation $y_1 - \lambda y_2 = 0$ embeds the system into a companion form, yielding the linear pencil

$$\left(\begin{bmatrix} D & K \\ I_n & 0 \end{bmatrix} - \lambda \begin{bmatrix} -M & 0 \\ 0 & I_n \end{bmatrix} \right) \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 0.$$

The resulting pencil has dimensions $(m + n) \times 2n$.

Rational Barycentric linearization. Standard linearizations of the AAA interpolant [19] typically convert the approximation into Newton or rational Krylov forms [30, 16] before constructing a companion pencil. Because our minimum perturbation method operates directly on the combined matrix $[\mathcal{B}, \mathcal{M}]$, we can instead linearize the barycentric representation directly, even though the resulting $\mathcal{B}$ is structurally singular. Consider the rational barycentric representation

$$R_b(\lambda) = \frac{\sum_{j=0}^{d} \dfrac{\omega_j F(z_j)}{\lambda - z_j}}{\sum_{j=0}^{d} \dfrac{\omega_j}{\lambda - z_j}}.$$

Writing $D_j = F(z_j)$ for the sampled coefficient matrices, define the block vector $y = [x^\top, y^{(0)\top}, y^{(1)\top}, \dots, y^{(d)\top}]^\top \in \mathbb{C}^{(d+2)n}$, where each block admits

$$y^{(j)} = \frac{1}{\lambda - z_j} x.$$

The singularity condition $R_b(\lambda)x = 0$ can then be directly embedded into the rectangular linear pencil $(\mathcal{M} - \lambda\mathcal{B})y = 0$, where $\mathcal{M}, \mathcal{B} \in \mathbb{C}^{(m+(d+1)n)\times(d+2)n}$ and

$$\mathcal{M} = \begin{bmatrix} 0 & \omega_0 D_0 & \omega_1 D_1 & \cdots & \omega_d D_d \\ I & z_0 I & 0 & \cdots & 0 \\ I & 0 & z_1 I & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ I & 0 & 0 & \cdots & z_d I \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & I & 0 & \cdots & 0 \\ 0 & 0 & I & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & I \end{bmatrix}.$$

Rational Newton linearization. Following the conversion framework in [16], we next construct the associated rectangular linearization for the rational Newton form $R_n(\lambda)$ obtained in Section 3.2. To derive the companion form, we seek a scalar parameter λ and a nonzero vector x such that $R_n(\lambda)x = 0$. To exploit the recurrence structure, we separate the highest-order term from the summation to write

$$R_n(\lambda)x = \sum_{j=0}^{d-1} \eta_j(\lambda) D_j x + \eta_d(\lambda) D_d x.$$

By applying the recurrence relation for the basis function $\eta_d(\lambda)$, we can eliminate the highest-order basis term. Substituting this recurrence into the equation yields

$$\beta_d(h_d - k_d\lambda) \sum_{j=0}^{d-1} \eta_j(\lambda) D_j x + (\lambda - \sigma_{d-1})\eta_{d-1}(\lambda) D_d x = 0. \tag{15}$$

After dividing by the scaling factor β_d, this condition is expressed entirely in terms of the d lower-order auxiliary vectors $\eta_0(\lambda)x, \dots, \eta_{d-1}(\lambda)x$. For $j = 1, \dots, d-1$, the Newton recurrence is equivalent to

$$\lambda(\eta_{j-1}(\lambda) + \beta_j k_j \eta_j(\lambda)) = \sigma_{j-1}\eta_{j-1}(\lambda) + \beta_j h_j \eta_j(\lambda).$$

Define the block vector

$$y = \begin{bmatrix} \eta_0(\lambda)x \\ \eta_1(\lambda)x \\ \vdots \\ \eta_{d-1}(\lambda)x \end{bmatrix} \in \mathbb{C}^{dn}, \tag{16}$$

Combining the $d-1$ recurrence equations with the eliminated relation (15) gives the rectangular linear pencil

$$(\mathcal{M} - \lambda\mathcal{B})y = 0,$$

where $\mathcal{M}, \mathcal{B} \in \mathbb{C}^{(m+(d-1)n)\times dn}$. More explicitly,

$$\mathcal{M} = \begin{bmatrix} h_d D_0 & h_d D_1 & \cdots & h_d D_{d-2} & h_d D_{d-1} - \frac{\sigma_{d-1}}{\beta_d} D_d \\ \sigma_0 I_n & \beta_1 h_1 I_n & & & \\ & \sigma_1 I_n & \beta_2 h_2 I_n & & \\ & & \ddots & \ddots & \\ & & & \sigma_{d-2} I_n & \beta_{d-1} h_{d-1} I_n \end{bmatrix},$$

and

$$\mathcal{B} = \begin{bmatrix} k_d D_0 & k_d D_1 & \cdots & k_d D_{d-2} & k_d D_{d-1} - \frac{1}{\beta_d} D_d \\ I_n & \beta_1 k_1 I_n & & & \\ & I_n & \beta_2 k_2 I_n & & \\ & & \ddots & \ddots & \\ & & & I_n & \beta_{d-1} k_{d-1} I_n \end{bmatrix}.$$

Here I_n denotes the $n \times n$ identity matrix. Therefore, the nonlinear problem $R_n(\lambda)x = 0$ is transformed into the singularity condition of a larger rectangular linear pencil and can be treated by the minimum perturbation method introduced in Section 3.1.

Eigenvector recovery. Once the rectangular linear pencil $(\mathcal{M} - \lambda\mathcal{B})y = 0$ is solved, the concatenated block vector y can be used to reconstruct the approximate eigenvector of the original problem. In all aforementioned representations, y admits a common structural form

$$y = \begin{bmatrix} \alpha_0(\lambda)x \\ \alpha_1(\lambda)x \\ \vdots \\ \alpha_{d-1}(\lambda)x \end{bmatrix},$$

where $x \in \mathbb{C}^n$. For square companion linearizations, the eigenvector recovery property [31] guarantees that the physical eigenvector x can be read off from a single block of the linearized eigenvector. In the rectangular case, the pencil usually does not admit exact eigenpairs, and the minimum perturbation solution breaks this strict block collinearity.

Nevertheless, a good candidate $\hat{\lambda}$ should produce a small residual on the finite representation, and the recurrence rows of the companion pencil then force the corresponding eigenvector to deviate only moderately from the strict block structure. We propose to exploit this structure as a diagnostic tool. While the true eigenvector must ultimately be computed via the singular value decomposition of $G(\lambda)$ for a validated λ, doing so for every candidate is computationally prohibitive. Instead, we construct a matrix U whose columns form an orthonormal basis for $\operatorname{span}\{\hat{y}^{(0)}, \hat{y}^{(1)}, \dots, \hat{y}^{(d-1)}\}$ and solve the dimensionally reduced minimization problem

$$\min_{\|c\|_2=1} \|G(\lambda)Uc\|_2$$

for $c \in \mathbb{C}^d$. The resulting vector $\tilde{x} = Uc$ provides an inexpensive proxy for the true residual, identifying the most promising candidates before committing to the expensive SVD evaluation on the original nonlinear problem.

To summarize, the full pipeline for a nonlinear rectangular problem consists of three stages: (i) rational approximation of the matrix-valued function via `sketchAAA`, (ii) linearization of the resulting finite representation into a rectangular companion pencil, and (iii) minimum perturbation extraction of the eigenpairs from this enlarged pencil, with block-structured eigenvector recovery used as a fast diagnostic evaluation. The following section examines the practical behavior of this procedure and its individual stages through three numerical experiments.

4 Applications and Numerical Examples

This section applies the methods introduced in Section 3 to three problems. Section 4.1 tests the full pipeline on a Laplace eigenvalue problem over polygonal domains, where the method of particular solutions produces a genuinely nonlinear rectangular problem. Section 4.2 considers a natively quadratic rectangular problem from a damped wave equation, thereby bypassing the rational approximation stage and testing the linearization and extraction steps in isolation. Section 4.3 examines whether randomized sketching can accelerate the minimum perturbation extraction.

All numerical experiments in this section are performed in MATLAB R2023b on an Apple M2 Pro processor. Code and documentation are available at the project repository[1].

[1] https://github.com/sucharush/revp

4.1 Laplace eigenvalue problem on polygons

We consider the Dirichlet Laplace eigenvalue problem on a bounded planar domain $\mathcal{P}$:

$$-\Delta u = \lambda u \quad \text{in } \mathcal{P}, \qquad u = 0 \quad \text{on } \partial\mathcal{P}. \tag{17}$$

For certain geometries (e.g. rectangles), separation of variables yields closed-form eigenpairs. For general polygons, solutions are not globally separable and the local regularity near corners becomes the dominant structural feature.

Singular corners Let $p \in \partial\mathcal{P}$ be a polygon vertex with interior angle $\omega \in (0, 2\pi)$, and introduce polar coordinates (r, θ) centered at p so that the two adjacent edges correspond to $\theta = 0$ and $\theta = \omega$. A classical corner expansion for solutions of elliptic boundary value problems [27, Theorem 3.4] implies that any nontrivial Dirichlet eigenfunction has the leading-order behavior

$$u(r, \theta) = \gamma\, r^{\alpha} \sin(\alpha\theta) + o(r^{\alpha}), \qquad \alpha = \frac{\pi}{\omega},\ \gamma \neq 0, \qquad r \to 0.$$

In particular, $\partial_r^m u(r, \theta) \sim C_m\, r^{\alpha - m} \sin(\alpha\theta)$, so if $\alpha \notin \mathbb{N}$ then choosing an integer $m > \alpha$ yields $\partial_r^m u$ unbounded as $r \to 0$. Thus u cannot be C^∞ at such a corner. At a straight boundary segment, a solution of $\Delta u + \lambda u = 0$ with homogeneous Dirichlet data extends across the segment by odd reflection [9, p. 395]. If a polygon corner has interior angle $\omega = \pi/k$ with $k \in \mathbb{N}$, iterating reflections across the two incident edges yields a smooth extension to a full neighborhood of the vertex. Otherwise $\alpha = \pi/\omega \notin \mathbb{N}$ and the local expansion contains fractional powers r^{α}, so analytic continuation around the corner is multi-valued. We call corners with $\alpha = \pi/\omega \notin \mathbb{N}$ singular and all others regular.

Following [3], the trial space is built from Fourier–Bessel functions derived on an unbounded wedge. For each singular corner p_j, one introduces local polar coordinates (r_j, θ_j) centered at p_j so that the adjacent edges correspond to $\theta_j = 0$ and $\theta_j = \omega_j$. Solving $(-\Delta - \lambda)u = 0$ on the corresponding wedge yields an explicit family of Fourier–Bessel functions encoding the correct corner behavior. Collecting these families over all singular vertices produces a global set of basis functions on $\mathcal{P}$ (each satisfies $(-\Delta - \lambda)u = 0$ in $\mathcal{P}$ away from the vertices).

Fourier–Bessel basis on a wedge We recall the construction of the Fourier–Bessel basis used in [3]. Fix a singular corner with parameter $\alpha > 0$ and consider the wedge

$$W_\alpha = \{(r, \theta) : r > 0,\ 0 < \theta < \pi/\alpha\} \subset \mathbb{R}^2,$$

whose boundary consists of the two rays $\theta = 0$ and $\theta = \pi/\alpha$. Here (r, θ) denote polar coordinates with respect to the corner point. Figure 1 illustrates the local wedge used to derive the basis functions.

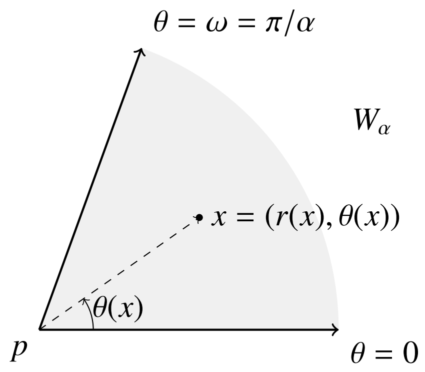


Figure 1: Local wedge near a polygon singular corner. The two rays represent the adjacent boundary edges, and the opening angle satisfies $\omega = \pi/\alpha$, $\alpha \notin \mathbb{N}$.

In polar coordinates (r, θ), the Laplacian is

$$\Delta u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}.$$

Hence the PDE becomes

$$-\left(u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}\right) = \lambda u.$$

Assume a separated form $u(r, \theta) = R(r)\,\Theta(\theta)$ and substitute it into the PDE

$$-r^2\frac{R''}{R} - r\frac{R'}{R} - \frac{\Theta''}{\Theta} = \lambda r^2.$$

Rearrange to separate r and θ:

$$-\frac{\Theta''(\theta)}{\Theta(\theta)} = \lambda r^2 + r^2\frac{R''(r)}{R(r)} + r\frac{R'(r)}{R(r)}.$$

The left-hand side depends only on θ, the right-hand side only on r, so both must equal a constant. Denote it by $\mu^2 \geq 0$:

$$-\frac{\Theta''(\theta)}{\Theta(\theta)} = \mu^2, \qquad \lambda r^2 + r^2\frac{R''(r)}{R(r)} + r\frac{R'(r)}{R(r)} = \mu^2.$$

The angular equation is

$$\Theta''(\theta) + \mu^2\Theta(\theta) = 0, \qquad \Theta(0) = 0,\ \Theta\left(\frac{\pi}{\alpha}\right) = 0.$$

A nontrivial solution exists only if $\mu = \mu_l^{(\alpha)} = l\alpha$ for $l = 1, 2, 3, \ldots$, in which case

$$\Theta_l^{(\alpha)}(\theta) = \sin(\mu_l^{(\alpha)}\theta) = \sin(\alpha\, l\theta).$$

The radial equation is $r^2R''(r) + rR'(r) + (\lambda r^2 - \mu^2)R(r) = 0$. Let $\kappa = \sqrt{\lambda}$ (for $\lambda > 0$) and set $z = \kappa r$. Then

$$z^2R''(z) + zR'(z) + (z^2 - \mu^2)R(z) = 0,$$

which is Bessel's equation of order μ. Its independent solutions are J_μ and Y_μ. If the wedge includes the origin and we require boundedness at $r = 0$, we retain only J_μ:

$$R_l^{(\alpha)}(\lambda; r) = J_{\mu_l^{(\alpha)}}(\sqrt{\lambda}\, r) = J_{\alpha l}(\sqrt{\lambda}\, r).$$

Combining radial and angular parts yields the α-dependent wedge family

$$\varphi_l^{(\alpha)}(\lambda; r, \theta) = J_{\alpha l}(\sqrt{\lambda}\, r)\ \sin(\alpha l \theta), \qquad l = 1, 2, 3, \ldots.$$

For any point expressed in polar coordinates (r, θ) centered at the corner with $r > 0$, the function $\varphi_l^{(\alpha)}(\lambda; r, \theta)$ satisfies $(-\Delta - \lambda)\varphi_l^{(\alpha)}(\lambda; \cdot) = 0$ pointwise. This does not depend on any global wedge geometry, the wedge derivation serves only to encode the correct corner singular structure.

Modified MPS algorithm We now describe the modified method of particular solutions (MPS) for the Laplace eigenvalue problem on polygons, following the formulation of [3]. Let $P \subset \mathbb{R}^2$ be a polygon with singular corners $\{p_i\}_{i=1}^I$ and interior angles $\{\omega_i\}_{i=1}^I$. For each corner p_i, set $\alpha_i = \pi/\omega_i$ and introduce local polar coordinates $(r_i(x), \theta_i(x))$ of a point $x \in P$ with respect to p_i. For a trial parameter λ, we approximate the solution by

$$u^{(n)}(\lambda, x) = \sum_{i=1}^{I} \sum_{l=1}^{n_i} c_{i,l}\, \varphi_l^{(\alpha_i)}(\lambda;\ r_i(x), \theta_i(x)), \tag{18}$$

where n_i is the number of basis functions associated with corner p_i and $n = \sum_{i=1}^{I} n_i$.

To impose the boundary condition $u = 0$ on ∂P, we sample boundary points $\{x_j\}_{j=1}^m \subset \partial P$ and enforce $u^{(n)}(\lambda, x_j) \approx 0$. Define the boundary sampling matrix as a block concatenation

$$A_B(\lambda) = \left[\, A_{B,1}(\lambda)\ \ A_{B,2}(\lambda)\ \ \cdots\ \ A_{B,I}(\lambda) \,\right] \in \mathbb{R}^{m \times n},$$

where each block $A_{B,i}(\lambda) \in \mathbb{R}^{m \times n_i}$ is given entrywise by

$$\left(A_{B,i}(\lambda)\right)_{j,l} = \varphi_l^{(\alpha_i)}(\lambda;\ r_i(x_j), \theta_i(x_j)), \qquad j = 1, \ldots, m,\ \ l = 1, \ldots, n_i.$$

Let $c_i = (c_{i,1}, \ldots, c_{i,n_i})^\top \in \mathbb{R}^{n_i}$ and $c = (c_1^\top, \ldots, c_I^\top)^\top \in \mathbb{R}^n$. Therefore, enforcing the boundary condition at the sampled nodes $\{x_j\}_{j=1}^m$ amounts to

$$\left(A_B(\lambda)c\right)_j = u^{(n)}(\lambda, x_j) \approx 0, \qquad j = 1, \ldots, m,$$

which is equivalently written as $A_B(\lambda)c \approx 0$.

However, when n is large, the boundary-only collocation matrix tends to become numerically rank-deficient, since the sampled basis functions may be nearly linearly dependent. For many values of λ one can choose c such that $u^{(n)}$ is close to the zero function on all sampled points, hence $A_B(\lambda)c \approx 0$ can occur even when λ is not close to an eigenvalue.

To rule out such near-zero solutions, we also sample m_I interior points $\{x_j\}_{j=1}^{m_I} \subset P \backslash \partial P$

and evaluate $u^{(n)}$ there. Define the interior sampling matrix $A_I(\lambda) \in \mathbb{R}^{m_I \times n}$ by the block concatenation

$$A_I(\lambda) = [\, A_{I,1}(\lambda) \;\; A_{I,2}(\lambda) \;\; \cdots \;\; A_{I,I}(\lambda) \,],$$

where each block $A_{I,i}(\lambda) \in \mathbb{R}^{m_I \times n_i}$ is given entrywise by $(A_{I,i}(\lambda))_{j,l} = \varphi_l^{(\alpha_i)}(\lambda;\, r_i(x_j), \theta_i(x_j))$. We then stack boundary and interior samples as

$$A(\lambda) = \begin{bmatrix} A_B(\lambda) \\ A_I(\lambda) \end{bmatrix} \in \mathbb{R}^{(m+m_I)\times n}.$$

Let $\mathcal{A}(\lambda) = \mathrm{range}(A(\lambda)) \subset \mathbb{R}^{(m+m_I)}$ be the sampled trial space. By construction, for any $c \in \mathbb{R}^n$ we have $u = A(\lambda)c \in \mathcal{A}(\lambda)$. To enforce the Dirichlet condition on the boundary without admitting spurious solutions, we look for $c \neq 0$ such that $A_B(\lambda)c \approx 0$ while $A_I(\lambda)c$ is not simultaneously small. Equivalently, we seek a nonzero vector $u = [u_B^\top, u_I^\top]^\top \in \mathcal{A}(\lambda)$ whose boundary block is small but interior block is nontrivial.

Compute a thin QR factorization

$$A(\lambda) = Q(\lambda)R(\lambda) = \begin{bmatrix} Q_B(\lambda) \\ Q_I(\lambda) \end{bmatrix} R(\lambda),$$

so the columns of $Q(\lambda)$ form an orthonormal basis of $\mathcal{A}(\lambda)$. Hence any $u \in \mathcal{A}(\lambda)$ with $\|u\|_2 = 1$ can be written as $u = Q(\lambda)v$ for some unit vector $v \in \mathbb{R}^n$, and we may partition

$$u = Q(\lambda)v = \begin{bmatrix} Q_B(\lambda)v \\ Q_I(\lambda)v \end{bmatrix} = \begin{bmatrix} u_B \\ u_I \end{bmatrix}. \tag{19}$$

Thus, minimizing the boundary block u_B reduces to $\min_{\|v\|_2=1} \|Q_B(\lambda)v\|_2$. Define the minimizer $v_* \in \mathbb{R}^n$ and set

$$\sigma(\lambda) := \|Q_B(\lambda)v_*\|_2 = \min_{\|v\|_2=1} \|Q_B(\lambda)v\|_2 = \sigma_{\min}(Q_B(\lambda)).$$

Let $u_* := Q(\lambda)v_*$ and partition $u_* = (u_{B*}, u_{I*})^\top$ with $u_{B*} = Q_B(\lambda)v_*$ and $u_{I*} = Q_I(\lambda)v_*$, then

$$1 = \|u_*\|_2^2 = \left\| \begin{bmatrix} Q_B(\lambda)v_* \\ Q_I(\lambda)v_* \end{bmatrix} \right\|_2^2 = \|Q_B(\lambda)v_*\|_2^2 + \|Q_I(\lambda)v_*\|_2^2 = \sigma(\lambda)^2 + \|u_{I*}\|_2^2. \tag{20}$$

Therefore, $\sigma(\lambda) = \sigma_{\min}(Q_B(\lambda))$ measures the smallest boundary residual attainable by unit-norm sampled vectors in $\mathcal{A}(\lambda)$. By (20), if $\sigma(\lambda)$ is close to zero, there exists a unit vector v such that $u = Q(\lambda)v(\lambda)$ satisfies $\|u_B\|_2 = \|Q_B(\lambda)v\|_2 \approx 0$ while the associated interior block remains nontrivial, so such λ are treated as admissible eigenvalue candidates.

The complete procedure for generating $Q_B(\lambda)$ at a given trial parameter is summarized in Algorithm 2.

The Laplace eigenvalue problem is thereby converted into a rectangular eigenvalue problem for $Q_B(\lambda)$. In the following, we solve it via two strategies: a direct scan-refinement

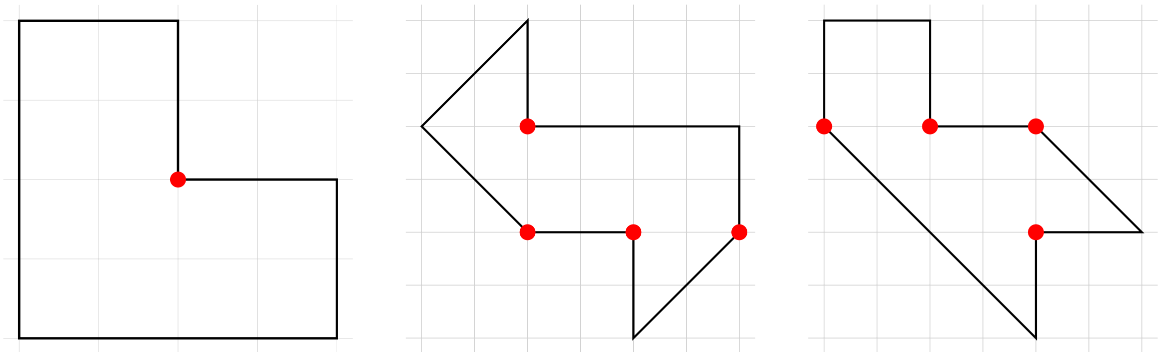
Algorithm 2 Boundary block $Q_B(\lambda)$ from Fourier–Bessel collocation

Input: Trial parameter λ, boundary points $\{x_j\}_{j=1}^{m} \subset \partial P$, interior points $\{x_j\}_{j=1}^{m_I} \subset P \setminus \partial P$
Output: Boundary block $Q_B(\lambda) \in \mathbb{R}^{m \times r}$
for each singular corner p_i, $i = 1, \dots, I$ **do**
    Evaluate the Fourier–Bessel basis at all sample points to form $A_{B,i}(\lambda) \in \mathbb{R}^{m \times n_i}$ and $A_{I,i}(\lambda) \in \mathbb{R}^{m_I \times n_i}$
end for
Concatenate $A_B(\lambda) = [A_{B,1}, \dots, A_{B,I}]$ and $A_I(\lambda) = [A_{I,1}, \dots, A_{I,I}]$
Stack $A(\lambda) = [A_B(\lambda)^\top, A_I(\lambda)^\top]^\top \in \mathbb{R}^{(m+m_I)\times n}$
Compute a QR factorization of $A(\lambda)$, with column pivoting and rank truncation if needed, and extract $Q_B(\lambda)$ from the boundary rows

procedure and a `sketchAAA`-based method that constructs a rational surrogate of $Q_B(\lambda)$ and extracts candidates from its linearization.

4.1.1 Experimental setup

We conducted numerical experiments for the Laplace eigenvalue problem using the modified MPS algorithm described in Section 4.1. The test set consists of the polygonal domains shown in Figures 2 and 3: an L-shaped domain, a pair of isospectral drums [14], and an H-shaped domain formed by two 2×2 squares connected by a horizontal channel of length 1 and width 0.3. The singular corners were selected according to the same criterion as in the previous section.



(a) L-shape (b) GWW1 (c) GWW2

Figure 2: L-shape and the GWW isospectral pair, with singular corners marked.

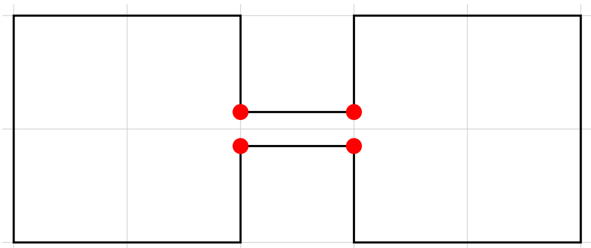


Figure 3: H-shape with channel width 0.3.

To define the discretization concisely, we introduce a base parameter t for each domain, which dictates both the number of expansion terms used at each singular corner and the number of Chebyshev sample points placed on each boundary edge. Following the baseline settings used in [3] whenever available, we set $t = 60$ for the L-shaped domain (one singular corner) and $t = 140$ for each drum domain (four singular corners each). For the H-shaped domain we use a coarser discretization with $t = 60$ (four singular corners), which is sufficient for the qualitative demonstration intended below.

The boundary of each domain was sampled edge by edge using these t Chebyshev points, while the interior was sampled using $m_I = 50$ uniformly distributed random points. For the H-shaped domain, we additionally enforced that at least 10% of the interior points lie inside the channel, so that this region is always represented in the interior sampling. For each trial value of λ, this formulation leads to real rectangular sampling matrices $A(\lambda)$ of size 410×60 for the L-shaped domain, 1170×560 for each drum domain, and 770×240 for the H-shaped domain. After QR factorization, the corresponding boundary blocks $Q_B(\lambda)$ remain rectangular, with sizes 360×60, 1120×560, and 720×240, respectively.

Later subsections occasionally adopt different discretization parameters to demonstrate specific phenomena, these deviations are stated where they occur.

4.1.2 Scan-refinement

Following [3], we constructed the basis using a column-pivoted QR factorization with a truncation threshold of 10^{-13}. Discarding columns associated with negligible diagonal entries of R stabilizes the process, mitigating spurious oscillations in $\sigma(\lambda)$ caused by nearly linearly dependent information in the Fourier–Bessel basis.

The eigenvalues were then approximated by searching for values of λ such that $\sigma(\lambda) = \sigma_{\min}(Q_B(\lambda)) \approx 0$. Each evaluation of $\sigma(\lambda)$ requires forming the sampling matrix $A(\lambda)$, computing its column-pivoted QR factorization with truncation, extracting the boundary block $Q_B(\lambda)$, and obtaining its smallest singular value via the SVD. The overall scan-refine procedure is summarized in Algorithm 3.

Algorithm 3 Scan-refine procedure

Input: Scan interval $[a, b]$; step size h; operator $Q_B(\cdot)$; local minimizer `minimize`; acceptance threshold τ

Output: Refined eigenvalue candidates $\{\hat{\lambda}_k\}$

Evaluate $\sigma_j \leftarrow \sigma_{\min}(Q_B(\lambda_j))$ on the uniform grid $\lambda_j = a + jh, \quad j = 0, 1, \dots, \lfloor (b-a)/h \rfloor$

for each local minimum $\sigma_j < \min(\sigma_{j-1}, \sigma_{j+1})$ with $\sigma_j < \tau$ **do**

  Set candidate interval $I_j \leftarrow [\lambda_{j-1}, \lambda_{j+1}]$

  $\hat{\lambda} \leftarrow$ `minimize`(σ, I_j) ▷ refine within I_j

end for

return accepted candidates $\{\hat{\lambda}_k\}$

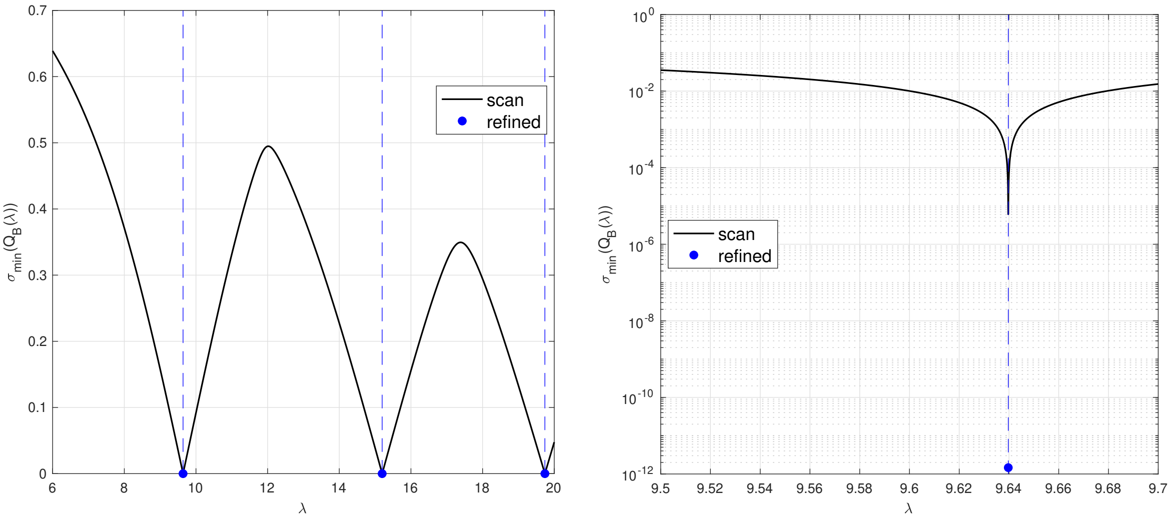
Since $m_I < m$ and $r \le n < m$ in all our experiments, the cost per evaluation of $\sigma(\lambda)$ is $O(mn^2)$, dominated by the column-pivoted QR factorization and the SVD. The total scan cost is proportional to $\lfloor \frac{b-a}{h} \rfloor$ such evaluations, and each subsequent local refinement adds n_{eval} further evaluations per candidate. We compared the following local refinement methods:

- **1D minimization.** We applied standard 1D search methods directly to $\sigma(\lambda)$: `fminsearch` (unconstrained, initialized at the interval midpoint), `fminbnd` (bounded), and golden section (bounded by construction). For all methods, `TolX` was set to 10^{-13}, with a maximum of 500 iterations.

- **`AAA`-based refinement.** Alternatively, we used the `AAA` algorithm introduced in Section 3.2 to build a rational surrogate of $1/\sigma(\lambda)$ on each candidate interval. Given sample points $Z = \{z_k\}$ and inverted objective values $R_k = 1/\sigma(z_k)$, the algorithm adaptively selects d support points to construct a type-$(d-1, d-1)$ barycentric rational approximant $r(z)$. The sample points were distributed on the real candidate interval or in a surrounding Bernstein ellipse with parameter $\rho = 1.05$. We set the maximum of `mmax` $= 100$ support points and relative fitting tolerance `tol` $= 10^{-13}$. The refined eigenvalue was then chosen among the candidate poles of r as the one giving the smallest value of σ.

We first illustrate the scan-refine procedure on the test domains using `fminsearch` with a uniform step size of $h = 0.025$ for the trial values of λ.

L-shaped domain. Figure 4 shows the resulting global scan of $\sigma(\lambda)$ for the L-shaped domain, alongside a zoomed-in view near the first detected minimum. Using `fminsearch` for the local refinement, the refined approximation for the fundamental eigenvalue is $\lambda_1 \approx 9.639723844021955$, which matches the reference value provided in [3] to all 14 reported decimal places.

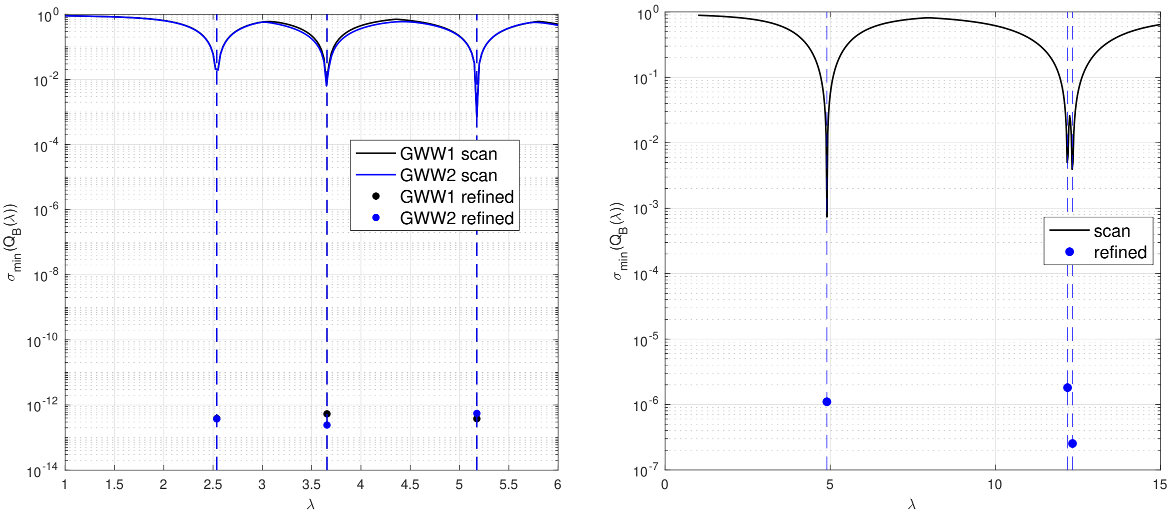


(a) Global scan of $\sigma(\lambda)$.

(b) Zoom near the first eigenvalue.

Figure 4: Results for the L-shaped domain, showing the detected λ and $\sigma_{\min}$.

Isospectral drums. For the pair of isospectral drums, Figure 5a demonstrates that the scan curves for both domains exhibit minima at nearly identical locations. The refined eigenvalues for the two shapes converge to the same values, fully agreeing with the results from [3] and remaining consistent with their isospectrality.



(a) $\sigma(\lambda)$ of the GWW drums. (b) $\sigma(\lambda)$ of the H shape with channel width 0.3.

Figure 5: Global scans of $\sigma(\lambda)$ for the pair of isospectral drums (left) and the H-shaped domain (right) with the detected λ and σ_{min}.

Comparison of local refinement methods. To evaluate the computational efficiency and numerical precision of the local refinement methods, we compared the final achieved minimum singular value (σ_{min}) and the number of SVD evaluations (n_{eval}) for each algorithm. Table 1 summarizes these metrics for the first eigenvalue of the L-shaped domain and the pair of isospectral drums (GWW1 and GWW2). The results for higher eigenvalues were strictly consistent with these trends.

Method	L-shape (λ_1)		GWW1 (λ_1)		GWW2 (λ_1)	
	σ_{min}	n_{eval}	σ_{min}	n_{eval}	σ_{min}	n_{eval}
`fminsearch`	1.46×10^{-12}	88	3.61×10^{-13}	84	3.56×10^{-13}	84
Golden section	1.46×10^{-12}	59	3.98×10^{-13}	59	3.67×10^{-13}	59
`fminbnd`	8.70×10^{-09}	17	2.48×10^{-08}	23	1.21×10^{-08}	20
AAA (Real)	4.88×10^{-06}	110	2.79×10^{-05}	145	2.35×10^{-05}	148
AAA (Ellipse)	1.10×10^{-04}	130	1.75×10^{-03}	130	1.56×10^{-03}	132

Table 1: Comparison of local refinement methods, targeting the first eigenvalue of the L-shaped domain ($\lambda_1 \approx 9.6397$) and the isospectral drums ($\lambda_1 \approx 2.5379$).

Both the unconstrained `fminsearch` and the bounded golden section search successfully resolved the minimizers to high precision, driving σ_{min} down to the 10^{-13} level. Notably, the golden section search outperformed `fminsearch` in efficiency, requiring just 59 objective evaluations compared to approximately 85. The golden section method also offers superior robustness: while the unconstrained `fminsearch` risks escaping the assigned candidate interval and converging to a neighboring root, the golden section search avoids this by strictly subdividing the provided bounds. In contrast, the built-in bounded `fminbnd` algorithm was fast but limited in precision, achieving an accuracy of only about 10^{-8}.

The AAA-based method was less competitive for this local refinement task. Despite requiring more than 100 evaluations per candidate interval, the resulting candidates only reduced $\sigma_{\min}$ to values between 10^{-3} and 10^{-6}. The method approximates $1/\sigma(\lambda)$ and then uses poles of the rational approximant as candidate locations. In our experiments, $1/\sigma(\lambda)$ forms a large finite peak near each minimizer, and the poles of the rational approximant provide only an indirect proxy for the location of this peak. This proxy was too coarse for high-precision local refinement in the tested examples.

H-shaped domain and clustered eigenvalues. We next apply the scan-refine procedure to the H-shaped domain with channel width 0.3 and $t = 60$. Because the horizontal connecting channel is narrow relative to the two 2×2 squares, the domain is close to two weakly coupled components. Accordingly, the detected minima in Figure 5b lie near the analytically known eigenvalues of a single square, $\lambda_{m,n} = \pi^2(m^2 + n^2)/4$ for $m, n \in \mathbb{N}^+$. The scan also reveals that each degenerate square eigenvalue is split into a closely spaced pair, reflecting the weak coupling through the channel.

In order to resolve the clustered eigenvalues visible in Figure 5b, we use golden section for the local refinement, ensuring that each search remains within its assigned candidate interval. With channel width 0.3 and step size $h = 0.025$, the splitting is large enough that each pair occupies separate candidate intervals, and golden section successfully resolves both members.

However, as the channel narrows further, the splitting decreases and will eventually fall below any fixed scan resolution. In that situation, two nearby eigenvalues are assigned to the same candidate interval, and even a bounded local minimizer will converge to only one of them. Refining the global grid is expensive, since each evaluation of $\sigma_{\min}(Q_B(\lambda))$ requires forming the sampled matrix, computing a QR factorization, and extracting the smallest singular value of the boundary block. Detecting such tightly clustered eigenvalues without refining the global grid requires a dedicated post-processing step.

To handle this situation, we add a post-processing step after the initial scan-refine procedure. For each accepted minimizer λ_*, we inspect the small singular values of $Q_B(\lambda_*)$. If only the smallest singular value is small, the candidate is treated as locally isolated. If several singular values are small, then $Q_B(\lambda_*)$ has an approximately higher-dimensional null space, signalling that the current dip may correspond to coupled modes. In that case, we perform an additional local search near λ_* at a finer resolution determined by the local sensitivity of $Q_B(\lambda)$.

Subdivision scale. The subdivision scale can then be chosen from Weyl's perturbation estimate. On the local window around λ_*, we fix the retained column set at the selection made at λ_* together with the sign convention of Section 2.3. Under this setting $Q_B(\lambda)$ inherits the smoothness of $A_r(\lambda)$, so the derivative $Q_B'(\lambda_*)$ below is well defined. For two nearby parameters λ and λ_*,

$$|\sigma_j(Q_B(\lambda)) - \sigma_j(Q_B(\lambda_*))| \le \|Q_B(\lambda) - Q_B(\lambda_*)\|_2.$$

Using the local approximation $Q_B(\lambda) = Q_B(\lambda_*) + (\lambda - \lambda_*)Q_B'(\lambda_*) + O(|\Delta\lambda|^2)$ and dropping the higher-order term, we obtain

$$|\sigma_j(Q_B(\lambda)) - \sigma_j(Q_B(\lambda_*))| \lesssim |\lambda - \lambda_*|\,\|Q_B'(\lambda_*)\|_2.$$

This suggests that resolving singular-value changes of size $|\sigma_j(Q_B(\lambda)) - \sigma_j(Q_B(\lambda_*))|$ requires a local step size on the order of

$$|\Delta\lambda| \gtrsim \frac{|\sigma_j(Q_B(\lambda)) - \sigma_j(Q_B(\lambda_*))|}{\|Q_B'(\lambda_*)\|_2}.$$

Since λ_* is already a refined rank-loss candidate, the smallest observed scale of the boundary residual is $\sigma_{\min}(Q_B(\lambda_*))$. We therefore use this value as a conservative scale for the singular-value change that should not be missed in the secondary scan. This gives the local resolution

$$\delta_{\text{fine}} = \gamma_{\text{fine}} \frac{\sigma_{\min}(Q_B(\lambda_*))}{\|Q_B'(\lambda_*)\|_2},$$

where $\gamma_{\text{fine}} < 1$ is a safety factor. Numerically, $\|Q_B'(\lambda_*)\|_2$ is estimated by a finite difference. Since the QR factor is not unique after column pivoting and truncation, we evaluate $Q_B(\lambda_* + h)$ using the same selected column set as at λ_*, and then fix the column signs consistently with $Q_B(\lambda_*)$, so that both factors are samples of the smooth representative above. The entire procedure is summarized in Algorithm 4.

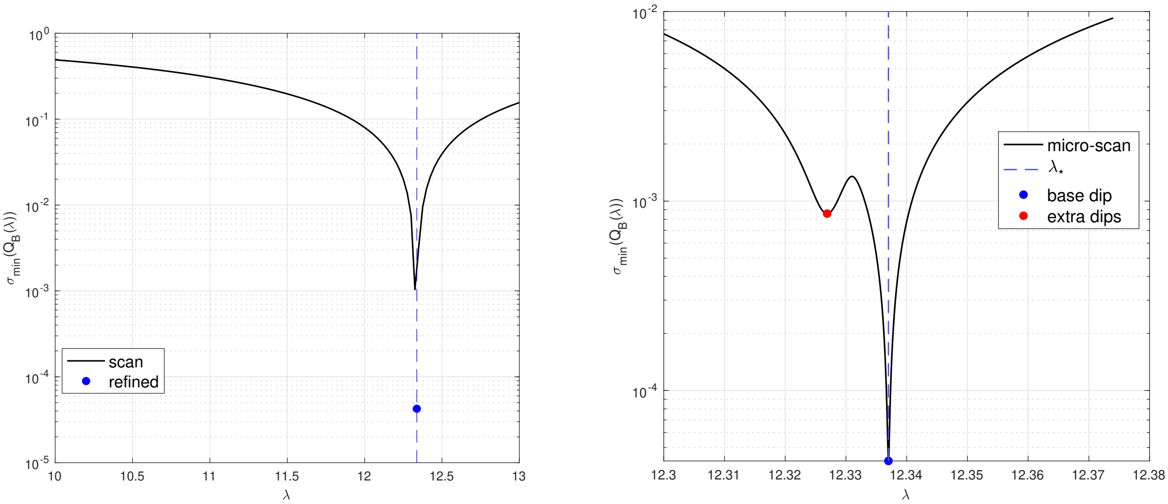
Algorithm 4 Cluster-aware post-processing of a refined candidate

1: **Input:** candidate λ_* with local interval $[a, b]$; operator $Q_B(\cdot)$; constants $\gamma_{\text{fine}} \in (0, 1)$, $\tau_{\text{clust}} > 1$; finite-difference step h
2: **Output:** refined candidates near λ_*
3: $s_* \leftarrow \sigma_{\min}(Q_B(\lambda_*))$
4: $k \leftarrow \#\{\, j : \sigma_j(Q_B(\lambda_*)) \le \tau_{\text{clust}}\, s_* \,\}$
5: **if** $k = 1$ **then return** $\{\lambda_*\}$ ▷ locally isolated
6: **end if**
7: $d_* \leftarrow \|Q_B(\lambda_* + h) - Q_B(\lambda_*)\|_2 / h$ ▷ frozen column set, matched signs
8: $\delta_{\text{fine}} \leftarrow \gamma_{\text{fine}}\, s_* / d_*$
9: $I_{\text{micro}} \leftarrow [\lambda_* - \delta,\ \lambda_* + \delta], \quad \delta = \max\{\lambda_* - a,\ b - \lambda_*\}$
10: Apply Algorithm 3 on I_{micro} with step size δ_{fine}
11: **return** the extra refined minima

For the H-shaped polygon with channel width 0.08, we scan the interval $[10, 13]$ with step size 0.025. In the subinterval $[12.30, 12.35]$ the scan-refine procedure accepts a single minimizer, $\lambda_* = 12.337000$, with $s_* = \sigma_{\min}(Q_B(\lambda_*)) = 4.25 \times 10^{-5}$ (Figure 6a). However, this candidate is not isolated. The four smallest singular values of $Q_B(\lambda_*)$ are 4.25×10^{-5},

5.44×10^{-5}, 3.32×10^{-3}, and 3.45×10^{-3}, so the second singular value lies within a factor 1.3 of s_* and the cluster test of Algorithm 4 fires. With $h = 10^{-4}$ and $\gamma_{\text{fine}} = 0.5$, the finite difference gives $d_* \approx 0.699$ and hence $\delta_{\text{fine}} \approx 3.041 \times 10^{-5}$. The resulting micro-scan (Figure 6b) reveals one additional, shallower dip at $\lambda = 12.326901$, where $\sigma_{\min}(Q_B(\lambda)) = 8.58 \times 10^{-4}$.

This behaviour is consistent with the geometry. In the limit of zero channel width the domain degenerates into two identical 2×2 squares, and each square eigenvalue $\lambda_{m,n} = \pi^2(m^2 + n^2)/4$ appears once per component. The candidate sits at $\lambda_{1,2} = \lambda_{2,1} = 5\pi^2/4 \approx 12.3370$, which is already double on each square and therefore carries multiplicity four in the disconnected limit. Opening a thin channel splits this fourfold eigenvalue, but at channel width 0.08 part of the splitting remains below the resolution of the current collocation. This explains why four singular values are small at λ_* while the micro-scan resolves only one additional minimum.



(a) Initial scan-refine result.

(b) Cluster-aware micro-scan.

Figure 6: Post-processing for a clustered candidate in the H-shaped domain with channel width 0.08.

Deflation and its limitations. After a rank-loss candidate λ_* has been found, with associated right singular vector v_* of the boundary block $Q_{B,r}(\lambda_*)$, an alternative idea is to deflate this direction and repeat the local search. However, such a deflation cannot be well-defined in a numerically stable way in the current example.

The first possibility would be to deflate directly in the orthogonalized coordinates of $Q_{B,r}(\lambda)$. This is problematic because these coordinates do not have a fixed meaning as λ varies. The matrix $Q_r(\lambda)$ is obtained from a dynamically truncated, column-pivoted QR factorization. Hence the selected columns, the numerical rank, and the signs or rotations of the orthogonal basis may change with λ. The frozen-basis device used before for estimating $\|Q_B'(\lambda_*)\|$ does not remove this issue: there it is only used to obtain a local finite-difference scale near λ_*, while the subsequent micro-scan is still performed on the original quantity $\sigma_{\min}(Q_B(\lambda))$. In contrast, deflation would use the frozen coordinates to define a new objective.

This would require a consistent transport of the deflated direction, which is not provided by the QR representation.

The alternative is to map the singular vector back to the original coefficient space. Let

$$A_r(\lambda_*) = A(\lambda_*)P_r(:, 1 : r) = Q_r(\lambda_*)R_r(\lambda_*),$$

where $P_r(:, 1 : r)$ contains the retained columns. If v_* is the right singular vector of the boundary block $Q_{B,r}(\lambda_*)$, then the corresponding coefficient vector in the original expansion is formally

$$c_* = P_r(:, 1 : r)R_r(\lambda_*)^{-1}v_*.$$

One could then form an isometry $V_\perp$ whose columns span the orthogonal complement of c_*, and replace the sampled matrix by

$$\widetilde{A}(\lambda) = A(\lambda)V_\perp.$$

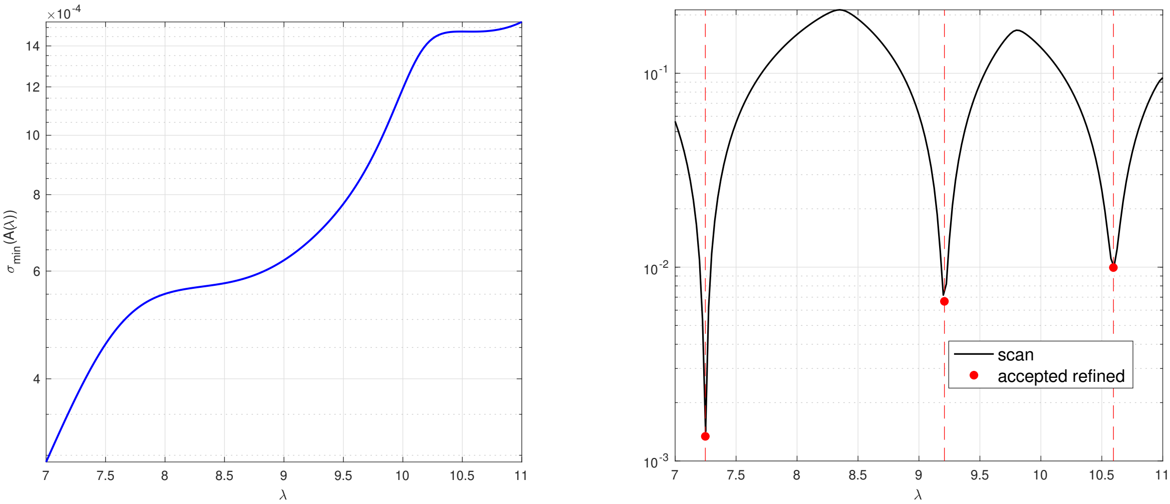
This removes one coefficient direction from the trial space.

In practice this construction is unreliable. As observed in the modified method of particular solutions, the underlying expansion basis can be severely ill-conditioned; the triangular factor $R_r(\lambda_*)$ may have a very large condition number. Although the triangular solve is backward stable, a large $\kappa(R_r)$ can produce a coefficient vector c_* with large forward error. In our computations, $\kappa(R_r)$ is often of order 10^{13} or larger. Therefore the recovered coefficient direction is not accurate enough to serve as a deflation direction. Moreover, coefficient-space orthogonality does not necessarily correspond to orthogonality of the represented physical functions, especially in an ill-conditioned basis. For these reasons we do not use deflation, and instead rely on a secondary local scan triggered by the presence of multiple small singular values.

4.1.3 SketchAAA

Having established that the scan-refine approach with direct $\sigma_{\min}$ evaluation is reliable, we now ask whether the cost of repeatedly evaluating $Q_B(\lambda)$ can be amortized by constructing a rational surrogate. To this end, we apply `sketchAAA` [19] to construct a rational surrogate of $Q_B(\lambda)$, and then linearize the resulting rational representation following the procedure introduced in Section 3.2.

We first record the reference behavior of the original sampled problem. We use the GWW1 drum discretization with $m_I = 50$ interior points and $t = 10$ for expansion terms and sample points per-edge, which gives $A(\lambda) \in \mathbb{R}^{130\times 40}$. As shown in Figure 7a, the smallest singular value of $A(\lambda)$ remains well above 10^{-4} on the sampled interval $[7, 11]$, so the matrix family is numerically full rank throughout this experiment. The reference scan of $\sigma_{\min}(Q_B(\lambda))$ is shown in Figure 7b, where three local minima are visible in the interval $[7, 11]$. These minima serve as the targets for the rational surrogate and the subsequent pencil-based extraction.



(a) $\sigma_{\min}(A(\lambda))$.

(b) $\sigma_{\min}(Q_B(\lambda))$.

Figure 7: Smallest singular values of $A(\lambda)$ and $Q_B(\lambda)$ over the sampled parameter range.

Rational surrogate and linearized extraction As discussed in Section 2.3, a smooth full-rank matrix family admits a smooth choice of QR basis, but independently computed numerical QR factors may still contain artificial gauge changes. Therefore, before applying `sketchAAA`, we align the sampled bases by an orthogonal Procrustes step introduced in Section 2.4. After computing $A(\lambda_k) = Q_k R_k$, we replace Q_k by $\tilde{Q}_k = Q_k \Omega_k$, where

$$\Omega_k = \arg\min_{\Omega^* \Omega = I} \|Q_k \Omega - Q_{k-1}\|_F.$$

This produces a more coherent representative of the sampled basis family.

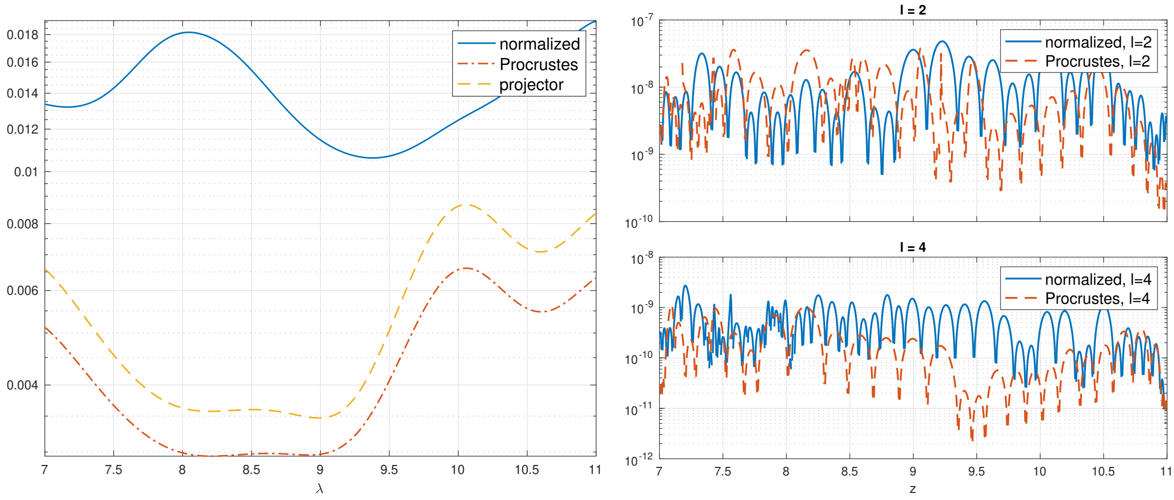
After alignment of the full QR basis $\tilde{Q}_k$, we extract its boundary block $\tilde{Q}_{B,k}$ and apply `sketchAAA` to the samples $\{\tilde{Q}_{B,k}\}$. As shown in Figure 8a, the successive difference $\|Q_{B,k} - Q_{B,k-1}\|_F$ becomes more regular and closer to the trend of the projector representation $Q_B(\lambda_k) Q_B(\lambda_k)^\top$.

Although $Q_B(\lambda) Q_B(\lambda)^\top$ is smoother as a subspace representation, it is singular for all λ. Linearizing a rational approximation of this projector would therefore lead to a singular matrix-valued problem, whose eigenvalue interpretation requires normal-rank language [24]. We therefore approximate the boundary basis representation directly, using either the normalized samples $Q_B(\lambda_k)$ or the Procrustes-aligned samples $Q_{B,k}$.

For each of these two sample families, `sketchAAA` selects support points from the sampled values and returns a barycentric rational approximant of $Q_B(\lambda)$ of the form (13). As shown in Figure 8b, the resulting approximation errors remain small for `tol` $= 10^{-11}$, reaching the 10^{-8} level for $\ell = 2$ and the 10^{-10} level for $\ell = 4$.

The detailed statistics are presented in Table 2. The Procrustes-aligned samples consistently lead to lower rational degrees and smaller approximation errors. This reduction in rational degree will directly reduce the size of the linearized pencils considered afterwards.

Having verified that the rational approximation error is small, we next examine whether the linearized pencils preserve the behavior of the smallest singular values of the original



(a) Consecutive change.

(b) Barycentric error.

Figure 8: Left: consecutive Frobenius changes of the raw QR basis, the Procrustes-aligned basis, and the projector representation. Right: relative barycentric approximation errors from `sketchAAA`.

		$\ell = 2$			$\ell = 4$		
`tol`	Method	Degree	Error	Time (s)	Degree	Error	Time (s)
10^{-11}	Normalized	41.6	1.45×10^{-8}	0.0229	52.6	5.92×10^{-10}	0.0474
10^{-11}	Procrustes	28.2	8.72×10^{-9}	0.0106	30.0	2.59×10^{-10}	0.0183
10^{-8}	Normalized	25.8	1.05×10^{-6}	0.00842	29.6	1.06×10^{-7}	0.0175
10^{-8}	Procrustes	18.0	8.41×10^{-7}	0.00481	21.0	2.91×10^{-8}	0.0107

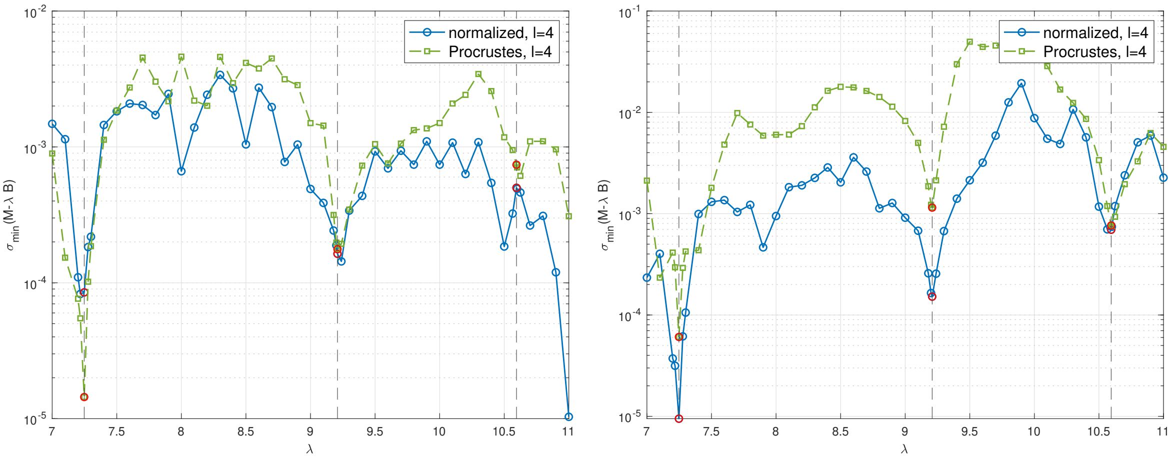
Table 2: Average `sketchAAA` approximation results over 5 runs.

matrix-valued problem. This is done by scanning the smallest singular value of the linearized pencil and comparing it with the reference curve in Figure 7b.

We consider two linearization strategies. The first converts the barycentric approximant into rational Newton form and applies the linearization described in Section 3.2. This is the standard route used in many rational linearization frameworks. The second keeps the barycentric form directly and builds the corresponding rectangular pencil without the Newton conversion. This second experiment is merely used to check whether possible failures are caused by the barycentric-to-Newton conversion or by the linearized rectangular extraction itself.

Figure 9 compares the smallest singular values of the rectangular pencils obtained from the Newton and barycentric linearizations. In both representations, the curves exhibit visible depressions near the three reference values obtained from the original scan of $\sigma_{\min}(Q_B(\lambda))$. This indicates that the rational approximation and the subsequent linearization preserve part of the small-singular-value landscape of the original problem.

The Procrustes-aligned variants generally give smoother curves than the independently normalized bases, consistent with the lower rational degrees reported in Table 2. The direct



(a) Newton form. (b) Barycentric form.

Figure 9: Smallest singular values of the linearized rectangular pencils. The dashed vertical lines and red markers indicate the reference values obtained by scan-refinement on the original sampled problem.

barycentric linearization also appears slightly cleaner than the Newton form. Nevertheless, all four curves contain additional dips and oscillations away from the reference locations. These suggest that solving the enlarged rectangular pencil may produce algebraic candidates that are not reliable approximations of the original problem.

Minimum perturbation extraction and post-filtering. We next apply the minimum perturbation method (MP) described in Section 3.1 to the rectangular pencils obtained from the rational surrogates. We use $\ell = 4$ and $\texttt{tol} = 10^{-11}$ throughout this comparison. The Procrustes-aligned surrogates lead to substantially smaller pencils: in Newton form, the normalized and Procrustes-aligned pencils have sizes 2320×2280 and 1240×1200 respectively; in barycentric form, the corresponding sizes are 2400×2360 and 1320×1280.

We first rank the computed MP eigenpairs by the relative residual of the linearized pencil, after filtering candidates by $\Re(\hat{\lambda}) \in [7, 11]$ and $|\Im(\hat{\lambda})| \leq 0.1$. Table 3 reports representative results for the Newton linearization. The residual-ranked candidates have pencil residuals close to machine precision, but their real parts remain far from the reference eigenvalues.

To obtain a more relevant diagnostic, we compute approximated eigenvectors of the surrogate matrix $R_b(\lambda)$ following Section 3.3. For each pencil eigenvector from minimum perturbation method, we construct a trial subspace U from its block directions and rank the candidate by the reduced residual $\rho_{\mathrm{red}}(\lambda) = \min_{\|c\|_2=1} \|R_b(\lambda)Uc\|_2$.

Table 4 reports the re-ranked candidates for the Procrustes-aligned Newton pencil. The reduced residual $\rho_{\mathrm{red}}(\lambda)$ gives a more meaningful ordering than the pencil residual, since candidates near the reference dips are promoted among the leading entries. However, several spurious or repeated candidates remain interleaved with them, and the separation is not strong enough to provide a reliable selection rule.

The preceding experiment still involves the conversion from barycentric to rational Newton form. To check whether the poor extraction is caused primarily by this conversion,

Basis	$\Re(\hat{\lambda})$	$\lvert\Im(\hat{\lambda})\rvert$	Pencil residual	$\lvert\hat{\lambda} - \lambda_*\rvert$
Normalized	8.0529	2.08×10^{-3}	1.13×10^{-16}	8.05×10^{-1}
	8.0529	2.08×10^{-3}	1.16×10^{-16}	8.05×10^{-1}
	7.4728	6.31×10^{-3}	1.16×10^{-16}	2.25×10^{-1}
	8.0529	2.08×10^{-3}	1.19×10^{-16}	8.05×10^{-1}
	8.0529	2.08×10^{-3}	1.21×10^{-16}	8.05×10^{-1}
Procrustes	7.1033	1.62×10^{-2}	2.84×10^{-16}	1.46×10^{-1}
	7.1033	1.62×10^{-2}	3.27×10^{-16}	1.46×10^{-1}
	7.1033	1.62×10^{-2}	3.29×10^{-16}	1.46×10^{-1}
	7.1033	1.62×10^{-2}	3.54×10^{-16}	1.46×10^{-1}
	7.1033	1.62×10^{-2}	3.69×10^{-16}	1.46×10^{-1}

Table 3: Representative MP candidates from the Newton linearization, ranked by the relative residual of the linearized pencil.

$\Re(\hat{\lambda})$	$\lvert\Im(\hat{\lambda})\rvert$	$\rho_{\text{red}}(\hat{\lambda})$	Pencil residual	Reference	$\lvert\hat{\lambda} - \lambda_*\rvert$
7.2473	2.39×10^{-4}	1.98×10^{-3}	1.07×10^{-7}	7.2479	6.67×10^{-4}
7.1033	1.62×10^{-2}	7.69×10^{-3}	9.48×10^{-15}	7.2479	1.46×10^{-1}
10.6682	5.56×10^{-3}	2.21×10^{-2}	1.06×10^{-6}	10.5950	7.34×10^{-2}
7.2036	1.06×10^{-2}	3.35×10^{-2}	2.28×10^{-6}	7.2479	4.56×10^{-2}
7.4160	3.43×10^{-2}	5.25×10^{-2}	6.83×10^{-16}	7.2479	1.72×10^{-1}
7.2154	9.56×10^{-2}	5.93×10^{-2}	4.79×10^{-7}	7.2479	1.01×10^{-1}
7.4160	3.43×10^{-2}	6.10×10^{-2}	3.98×10^{-14}	7.2479	1.72×10^{-1}
7.1033	1.62×10^{-2}	7.68×10^{-2}	8.31×10^{-16}	7.2479	1.46×10^{-1}
7.2147	1.95×10^{-3}	8.11×10^{-2}	2.50×10^{-6}	7.2479	3.33×10^{-2}
9.4659	8.45×10^{-2}	8.89×10^{-2}	1.63×10^{-7}	9.2090	2.70×10^{-1}

Table 4: Top 10 candidates with smallest residual $\rho_{\text{red}}(\hat{\lambda})$ from the Procrustes-aligned Newton pencil.

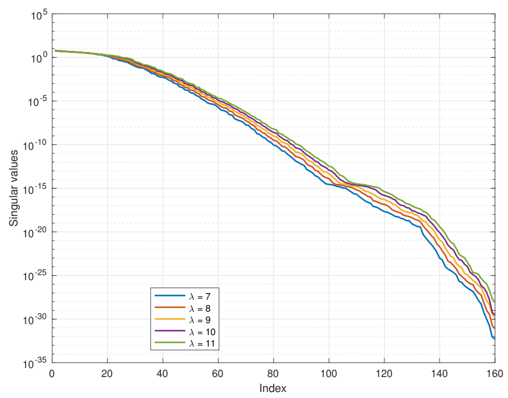
we repeat the MP extraction using the direct barycentric linearization introduced in Section 3.3. The same issue appears for the normalized barycentric pencil: many candidates have very small pencil residuals while remaining far from the reference values. For the Procrustes-aligned barycentric pencil, the near-real candidates are better separated, so we use the tighter prefilter $\lvert\Im(\hat{\lambda})\rvert \leq 10^{-2}$ instead of $\lvert\Im(\hat{\lambda})\rvert \leq 10^{-1}$. As shown in Table 5, only three candidates remain under this filter. Two of them are close to reference locations, but the middle reference value is still not accurately recovered.

$\Re(\hat{\lambda})$	$\lvert\Im(\hat{\lambda})\rvert$	$\rho_{\text{red}}(\hat{\lambda})$	Pencil residual	Reference	$\lvert\hat{\lambda} - \lambda_*\rvert$
7.2482	5.36×10^{-10}	1.41×10^{-3}	2.74×10^{-7}	7.2479	2.45×10^{-4}
10.5872	2.13×10^{-10}	1.04×10^{-2}	1.15×10^{-5}	10.5950	7.76×10^{-3}
8.9343	3.49×10^{-10}	7.96×10^{-2}	1.74×10^{-5}	9.2090	2.75×10^{-1}

Table 5: Top candidates with smallest residual $\rho_{\text{red}}(\hat{\lambda})$ from the Procrustes-aligned barycentric pencil.

Discussion on the limitations. The experiments above indicate that the failure of this approach is not caused by the rational approximation stage alone, but rather by intrinsic limitations of the sampled matrix family and the algebraic extraction process for this specific setup.

First, there is an inherent conflict between subspace smoothness and the strength of the eigenvalue signal, which is dictated by the collocation size. To obtain a smooth and consistently represented basis $Q_B(\lambda)$, we must use a relatively coarse collocation with $t = 10$ to avoid numerical rank loss. However, this smooth basis yields a weak discrete eigenvalue signal. As shown in Figure 7b, the local minima of $\sigma_{\min}(Q_B(\lambda))$ are only at the 10^{-3} level. If we attempt to recover a stronger signal by switching to a finer collocation with $t = 40$, the sampled matrix family becomes $A(\lambda) \in \mathbb{C}^{370\times 160}$. While this larger dimension allows the smallest singular value to achieve the 10^{-10} level, the collocation matrix becomes severely rank-deficient across the parameter interval, as illustrated in the left panel of Figure 10. In this regime, the underlying column space and the corresponding orthogonal projector $P(\lambda) = Q(\lambda)Q(\lambda)^\top = A(\lambda)A(\lambda)^+$ become highly ill-conditioned. Consequently, the consecutive basis changes fluctuate wildly even after Procrustes alignment, which is shown in the right panel of Figure 10. This loss of regularity is intrinsic to the matrix family itself and completely destroys the smoothness required to construct a valid rational surrogate.



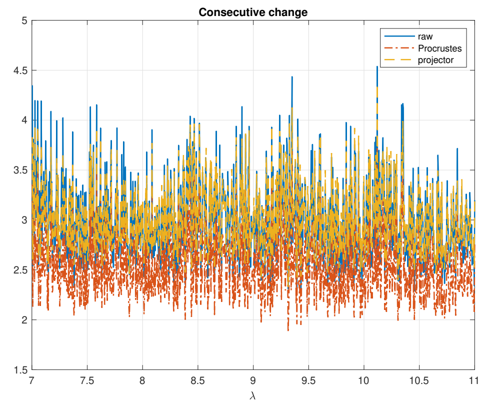


Figure 10: Left: singular values of $A(\lambda)$ at the representative values $\lambda = 7, 8, 9, 10, 11$, showing severe numerical rank deficiency throughout the interval. Right: consecutive Frobenius norm differences for the normalized QR basis, the Procrustes-aligned basis, and the projector representation.

Second, even when the rational approximation succeeds, such as in the case where $t = 10$, the subsequent MP extraction becomes numerically unreliable in this setting. In the final step, MP solves the square generalized eigenvalue problem $V_{21}^\top x = \lambda V_{11}^\top x$. Across our experiments, the matrix V_{11} is highly ill-conditioned, with $\mathrm{cond}(V_{11})$ well above 10^{11}. This makes the extracted eigenvalues extremely sensitive to numerical perturbations, inevitably producing inaccurate or spurious candidates.

Overall, these observations highlight a series of numerical obstacles when applying the rational linearization route to this particular collocation problem. To determine whether these difficulties originate in the rational approximation or in the extraction machinery itself, we turn in the next section to a problem where the collocation directly yields a quadratic matrix polynomial, bypassing the surrogate construction entirely.

4.2 Rectangular quadratic eigenvalue problem

The numerical example in Section 4.1.3 highlighted a significant challenge in our extraction pipeline. When the underlying problem exhibits highly nonlinear parameter dependence, constructing a valid rational surrogate with high degree is required to ensure the approximation accuracy. This inevitably generates large linearized companion pencils. Consequently, numerical errors accumulate from the initial rational approximation to SVD on the large, sparse companion-form matrix, and significantly degrade the quality of the final extraction.

This observation raises a natural question: if we isolate the extraction machinery from the rational approximation step, does the pipeline succeed? Specifically, if we consider a problem where the finite-dimensional collocation directly yields a natively quadratic matrix pencil, we can bypass the surrogate construction entirely and apply the rectangular linearization and MP extraction directly. To explore this, we turn our attention to a rectangular quadratic eigenvalue problem arising from a damped-wave equation.

We consider a damped-wave equation

$$u_{tt} + 2\alpha u_t - \beta u_{txx} - u_{xx} = 0$$

on a normalized domain with boundary condition $u(-1,t) = u(1,t) = 0$. Applying the modal ansatz $u(x,t) = e^{\lambda t} v(x)$ gives

$$\begin{cases} (\lambda^2 + 2\alpha\lambda)v(x) - (\beta\lambda + 1)v''(x) = 0, & x \in (-1,1), \\ v(-1) = v(1) = 0. \end{cases} \tag{21}$$

To approximate the solution, we define the basis functions

$$\varphi_j(x) = (1 - x^2)T_j(x), \qquad j = 0, \dots, n-1,$$

where $T_j(x)$ is the Chebyshev polynomial of the first kind of degree j. The factor $(1 - x^2)$ ensures that each basis function inherently satisfies the Dirichlet boundary conditions $\varphi_j(-1) = \varphi_j(1) = 0$. We approximate the eigenfunction by the expansion

$$v_n(x) = \sum_{j=0}^{n-1} c_j \varphi_j(x) = \Phi(x)c,$$

where $c = [c_0, \dots, c_{n-1}]^\top$ is the vector of unknown coefficients, and $\Phi(x) = [\varphi_0(x), \dots, \varphi_{n-1}(x)]$. Because the boundary conditions are built into the basis, the collocation equations only need to enforce the differential equation at a set of m interior collocation points $x_i \in (-1,1)$ for $i = 1, \dots, m$, where x_i are the interior Chebyshev points $x_i = -\cos(i\pi/(m+1))$. Substituting v_n into the differential equation in (21) and evaluating at each x_i yields the collocation equations

$$\sum_{j=0}^{n-1} \Big[(\lambda^2 + 2\alpha\lambda)\varphi_j(x_i) - (\beta\lambda + 1)\varphi_j''(x_i)\Big] c_j = 0.$$

To compute

$$\varphi_j''(x) = -2T_j(x) - 4xT_j'(x) + (1 - x^2)T_j''(x),$$

we recall that $T_j(x)$ satisfies the Chebyshev differential equation $(1 - x^2)T_j''(x) - xT_j'(x) + j^2T_j(x) = 0$. Substituting it yields

$$\varphi_j''(x) = -3xT_j'(x) - (j^2 + 2)T_j(x).$$

Using the relation $T_j'(x) = jU_{j-1}(x)$, where U_{j-1} is the Chebyshev polynomial of the second kind, we obtain the explicit formula

$$\varphi_j''(x) = -3jxU_{j-1}(x) - (j^2 + 2)T_j(x).$$

Thus, the collocation discretization results in the quadratic eigenvalue problem

$$A(\lambda)c = 0,$$

where $A(\lambda) \in \mathbb{C}^{m\times n}$ is defined by $[A(\lambda)]_{ij} = (\lambda^2 + 2\alpha\lambda)\varphi_j(x_i) - (\beta\lambda + 1)\varphi_j''(x_i)$ for $i = 1, \ldots, m$ and $j = 0, \ldots, n - 1$. Equivalently, this can be written as

$$A(\lambda) = \lambda^2 M + \lambda D + K,$$

where the mass, damping, and stiffness matrices are respectively defined as

$$[M]_{ij} = \varphi_j(x_i), \qquad [K]_{ij} = -\varphi_j''(x_i), \qquad D = 2\alpha M + \beta K.$$

The square collocation case corresponds to $m = n$, while the oversampled rectangular case uses $m > n$.

The damped-wave PDE is the continuous analogue of a classical damped second-order dynamic system. Upon spatial discretization, it naturally leads to the algebraic quadratic eigenvalue problem $(\lambda^2 M + \lambda D + K)c = 0$. Quadratic eigenvalue problems of this form are standard in damped vibrations and structural dynamics [21], and also occur frequently in dissipative acoustic vibration models [2].

The minimum perturbation method in Algorithm 1 applied to the rectangular linearization returns candidate eigenvalues $\hat{\lambda}$ and pencil eigenvectors $\hat{y} \in \mathbb{C}^{2n}$. Following the general eigenvector recovery framework described in Section 3.3, we avoid the expensive full SVD evaluation for every candidate. For each computed vector, we partition it into two blocks $\hat{y} = [\hat{y}_1^\top, \hat{y}_2^\top]^\top$ with $\hat{y}_1, \hat{y}_2 \in \mathbb{C}^n$. Since an exact eigenpair of the linearized pencil should satisfy $y_1 = \lambda y_2$, we construct our tailored trial subspace as $\mathcal{S}(\hat{\lambda}, \hat{y}) = \operatorname{span}\{\hat{y}_2, \hat{y}_1\}$. The approximate coefficient vector $\hat{c}$ is then efficiently extracted via the dimensionally reduced minimization problem

$$\hat{c} = \underset{\substack{c \in \mathcal{S}(\hat{\lambda}, \hat{y}) \\ \|c\|_2 = 1}}{\operatorname{argmin}} \|A(\hat{\lambda})c\|_2.$$

Finally, the candidate is accepted only if the norm-aware residual

$$\rho(\hat{\lambda}, \hat{c}) = \frac{\|A(\hat{\lambda})\hat{c}\|_2}{|\hat{\lambda}|^2\|M\hat{c}\|_2 + |\hat{\lambda}|\|D\hat{c}\|_2 + \|K\hat{c}\|_2}$$

is sufficiently small. Candidates failing this check are treated as spurious eigenpairs. The entire process is summarized in Algorithm 5.

Algorithm 5 Rectangular QEP extraction via linearization and minimum perturbation

Input: coefficient matrices $M, D, K \in \mathbb{C}^{m\times n}$; residual threshold τ
Output: accepted eigenpairs $\{(\hat{\lambda}_k, \hat{c}_k)\}$
Form the rectangular companion pencil $\mathcal{M} - \lambda\mathcal{B} \in \mathbb{C}^{(m+n)\times 2n}$ ▷ Section 3.3
Apply Algorithm 1 to obtain pencil eigenpairs $\{(\hat{\lambda}, \hat{y})\}$ ▷ $\hat{y} = [\hat{y}_1^\top, \hat{y}_2^\top]^\top$
for each candidate $(\hat{\lambda}, \hat{y})$ **do**
    Construct trial subspace $U \leftarrow \operatorname{orth}[\hat{y}_2,\ \hat{y}_1/\hat{\lambda}]$
    $\tilde{c} \leftarrow \arg\min_{\|c\|_2=1} \|A(\hat{\lambda})\,Uc\|_2, \quad \hat{c} \leftarrow U\tilde{c}$
    Compute norm-aware residual $\rho(\hat{\lambda}, \hat{c}) \leftarrow \|A(\hat{\lambda})\hat{c}\|_2/(|\hat{\lambda}|^2\|M\hat{c}\|_2 + |\hat{\lambda}|\|D\hat{c}\|_2 + \|K\hat{c}\|_2)$
    if $\rho(\hat{\lambda}, \hat{c}) \le \tau$ **then**
        Accept $(\hat{\lambda}, \hat{c})$
    end if
end for

Numerical Results In the following experiments, we set $\alpha = 3$ and $\beta = 0.04$. The analytical spatial modes for this boundary value problem are $v_k(x) = \sin(k\pi(x+1)/2)$ for $k \ge 1$, which naturally lead to the characteristic equation

$$\lambda^2 + (2\alpha + \beta\mu_k)\lambda + \mu_k = 0,$$

where $\mu_k = (k\pi/2)^2$. The roots of this equation serve as our exact ground truth.

As illustrated in Figure 11, the proposed extraction pipeline applied to a rectangular collocation is able to distinguish the true physical eigenvalues from spurious ones.

Table 6 compares the performance of the traditional square collocation ($m = n$) against the oversampled rectangular collocation ($m = n + 10$) for selected medium-to-high frequency modes. Low-frequency modes are excluded as both methods resolve them to near machine precision. The results demonstrate that our extraction pipeline can successfully process the rectangular pencil, accurately recovering the physically meaningful eigenvalues while filtering out spurious ones. While the solutions for the square case satisfy the linearized block structure exactly and thus yield small residuals, their forward error deteriorates significantly for higher modes. In contrast, the oversampled rectangular collocation inherently carries a larger residual but frequently yields a more accurate approximation of the exact physical eigenvalues.

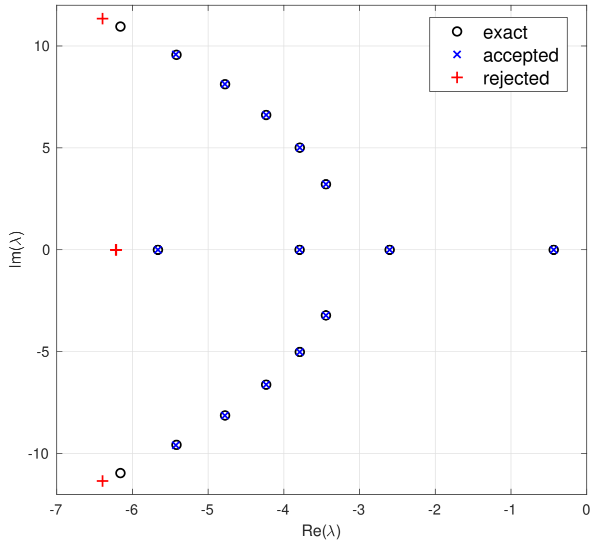


Figure 11: Minimum perturbation candidates for the rectangular collocation with $n = 15$ and $m = 25$. A candidate is accepted if its norm-aware residual $\rho(\hat{\lambda}, \hat{c}) \leq 10^{-2}$.

Exact Eigenvalue λ	Forward Error		Residual ρ	
	$\lvert\hat{\lambda}_{\text{sq}} - \lambda\rvert$	$\lvert\hat{\lambda}_{\text{rect}} - \lambda\rvert$	ρ_{sq}	ρ_{rect}
Degree $n = 30$				
$-5.418 + 9.568i$	8.119×10^{-14}	6.250×10^{-13}	7.119×10^{-13}	1.941×10^{-11}
$-6.158 + 10.954i$	2.023×10^{-12}	7.318×10^{-14}	1.143×10^{-12}	1.584×10^{-10}
$-6.997 + 12.284i$	1.372×10^{-10}	2.873×10^{-12}	5.485×10^{-13}	1.753×10^{-8}
$-7.935 + 13.557i$	1.573×10^{-9}	8.346×10^{-11}	9.604×10^{-13}	7.097×10^{-8}
$-8.971 + 14.767i$	3.498×10^{-8}	8.095×10^{-8}	3.436×10^{-13}	2.742×10^{-6}
Degree $n = 40$				
$-10.106 + 15.911i$	3.482×10^{-13}	2.063×10^{-13}	2.142×10^{-12}	2.208×10^{-11}
$-11.340 + 16.982i$	1.151×10^{-11}	4.298×10^{-12}	2.066×10^{-12}	1.443×10^{-9}
$-12.672 + 17.973i$	1.119×10^{-10}	1.118×10^{-12}	1.549×10^{-12}	4.988×10^{-9}
$-14.103 + 18.875i$	1.965×10^{-9}	4.600×10^{-10}	1.566×10^{-12}	1.600×10^{-7}
$-15.633 + 19.679i$	1.386×10^{-8}	2.884×10^{-9}	9.508×10^{-13}	4.075×10^{-7}

Table 6: Comparison of forward errors and norm-aware residuals $\rho(\hat{\lambda}, \hat{c})$ for selected medium-to-high frequency modes. A tolerance of $\rho \leq 10^{-5}$ is used for filtering spurious candidates in rectangular cases.

4.3 Sketched minimum perturbation for eigenvalue extraction

Section 4.2 provides an example where the minimum perturbation method is successfully applied to solve linearized rectangular pencils. However, as the dimension of the resulting pencil grows, the algorithm becomes increasingly expensive due to the cost of performing the SVD on the tall-and-skinny matrix. In this section, we explore whether we can reduce this computational cost via randomized sketching techniques while preserving the algorithmic accuracy.

To isolate the effects of sketching from the discretization errors of physical problems or collocation methods, we consider a synthetic setup. Let $A \in \mathbb{R}^{n\times n}$ be a symmetric matrix with an exact simple eigenpair (λ, v), and let $Q \in \mathbb{R}^{n\times r}$ be an orthonormal basis such that the principal angle $\measuredangle(Q, v) = \varepsilon$. Our goal is to extract this eigenpair from the rectangular pencil $AQ - \lambda Q$. From this trial space span(Q), we can extract a unit vector $\hat{v} \in \mathbb{R}^n$ satisfying

$$\hat{v} = Q\hat{y} = v\cos\varepsilon + w\sin\varepsilon,$$

where $v, w \in \mathbb{R}^n$, $y \in \mathbb{R}^r$, $\|\hat{y}\|_2 = \|v\|_2 = \|w\|_2 = 1$ and $w \perp v$. Because $Av = \lambda v$, this extracted vector yields an approximate eigenpair $(\lambda, \hat{v})$ with a small residual in the rectangular problem:

$$\|(AQ - \lambda Q)\hat{y}\|_2 = \|(A - \lambda I_n)\hat{v}\|_2 = \|(A - \lambda I_n)w\sin\varepsilon\|_2 \leq \|A - \lambda I_n\|_2\varepsilon.$$

By leveraging the Rayleigh quotient $\rho = \hat{v}^\top A\hat{v}$ of symmetric A, substituting our extracted vector yields

$$\rho = (v^\top\cos\varepsilon + w^\top\sin\varepsilon)A(v\cos\varepsilon + w\sin\varepsilon) = \lambda + w^\top(A - \lambda I)w\sin^2\varepsilon.$$

Thus the eigenvalue error is bounded by

$$|\rho - \lambda| \leq \|A - \lambda I\|_2\sin^2\varepsilon \leq \|A - \lambda I\|_2\varepsilon^2.$$

Consequently, after Rayleigh quotient refinement, an error $O(\varepsilon)$ in the eigenvector guarantees an error $O(\varepsilon^2)$ in the eigenvalue.

To reduce the computational cost of extracting this pair, we introduce a randomized sketching matrix $S \in \mathbb{R}^{l\times n}$ and consider the sketched rectangular pencil $S(AQ - \lambda Q)$. Let $C = [Q, AQ] \in \mathbb{R}^{n\times 2r}$ denote the augmented matrix used by the minimum perturbation method. If S is a γ-subspace embedding for the $2r$-dimensional subspace range(C), the sketched residual satisfies

$$\sigma_{\min}(S(AQ - \lambda Q)) \leq \|S(AQ - \lambda Q)\hat{y}\|_2 \leq \sqrt{1+\gamma}\,\|A - \lambda I_n\|_2\,\varepsilon.$$

Beyond the residual level, the target direction itself is preserved in the sketched trailing subspace.

Lemma 4.1 (Containment of the target direction)**.** *In the setting above, let $SC = \tilde{U}\tilde{\Sigma}\tilde{V}^\top$ be the thin SVD of the sketched augmented matrix, and let $\tilde{V}_2 \in \mathbb{R}^{2r\times r}$ denote its trailing r right singular vectors. Define the target direction $y_* = [-\lambda\hat{y}^\top, \hat{y}^\top]^\top \in \mathbb{R}^{2r}$. If $\sigma_r(C) > 0$, then*

$$\sin\measuredangle(y_*, \mathrm{range}(\tilde{V}_2)) \leq \sqrt{\frac{1+\gamma}{1-\gamma}}\frac{\|A - \lambda I_n\|_2\,\varepsilon}{\sigma_r(C)\sqrt{1+|\lambda|^2}}. \tag{22}$$

Proof. Applying Lemma 2.1 to the matrix SC with the vector y_* gives

$$\sin\measuredangle(y_*, \mathrm{range}(\tilde{V}_2)) \leq \frac{\|SC\,y_*\|_2}{\tilde{\sigma}_r\,\|y_*\|_2}.$$

The numerator satisfies $\|SCy_*\|_2 \le \sqrt{1+\gamma}\,\|Cy_*\|_2 = \sqrt{1+\gamma}\,\|(A-\lambda I_n)\hat{v}\|_2 \le \sqrt{1+\gamma}\,\|A-\lambda I_n\|_2\,\varepsilon$, and the embedding property yields $\tilde{\sigma}_r \ge \sqrt{1-\gamma}\,\sigma_r(C)$. Since $\|y_*\|_2 = \sqrt{1+|\lambda|^2}$, dividing gives the stated bound. □

The target direction therefore remains within $O(\varepsilon)$ of the sketched trailing subspace.

Experimental setup. We begin with a synthetic problem in which the target eigenpair and the quality of the trial subspace are controlled explicitly. Let $n = 256$, and construct a symmetric matrix $A \in \mathbb{R}^{n\times n}$ by

$$A = W\Lambda W^\top,$$

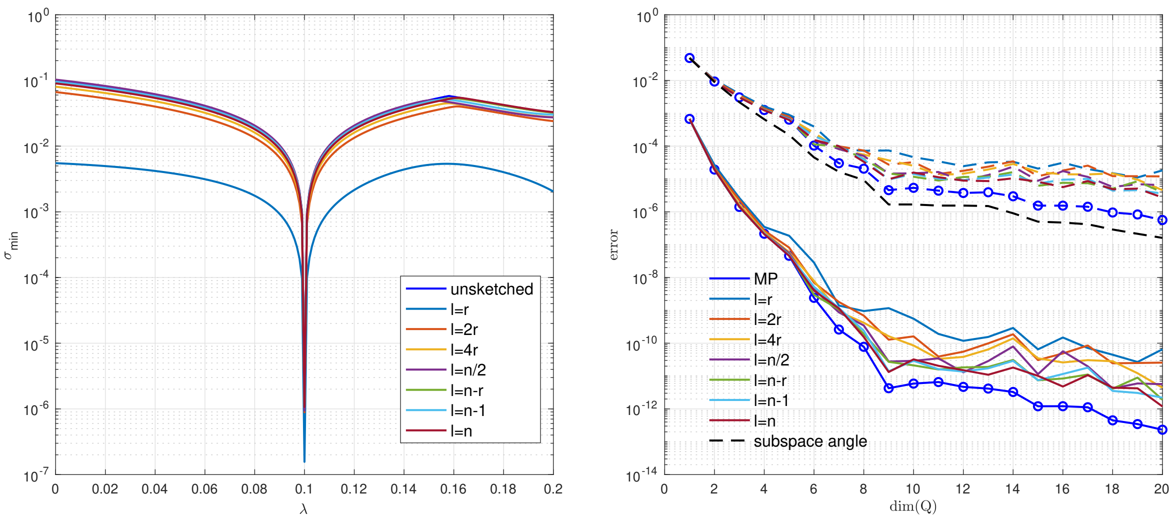
where $W \in \mathbb{R}^{n\times n}$ is a random orthogonal matrix. The diagonal matrix Λ contains the prescribed eigenvalues $-228, -227, \dots, -1, 0.1, 0.2, \dots, 2.8$. The reference eigenpair (λ, v) is chosen as the eigenpair of A associated with the smallest eigenvalue in magnitude, so that $\lambda = 0.1$.

To generate a sequence of trial subspaces, we draw a random symmetric matrix $E \in \mathbb{R}^{n\times n}$ scaled by 5×10^{-3} and define $A_k = A + kE$ for $k = 1, \dots, k_{\max}$, where $k_{\max} = 20$. For each k, we compute the eigenvector of A_k associated with the eigenvalue of smallest magnitude. Orthonormalizing the first k such vectors gives a trial matrix $Q_k \in \mathbb{R}^{n\times r_k}$, where $r_k \le k$. For the largest trial matrix, the angle to the reference eigenvector satisfies $\measuredangle(Q_{20}, v) \approx 1.6\times 10^{-7}$.

For each Q_k, we evaluate the rectangular matrix $AQ_k - \lambda Q_k$ and apply a randomized Gaussian sketching matrix $S \in \mathbb{R}^{l\times n}$ scaled by $1/\sqrt{l}$, with

$$\ell \in \{r_k,\ 2r_k,\ 4r_k,\ n/2,\ n-r_k,\ n-1,\ n\}.$$

All reported quantities are medians over 30 independent runs.



(a) $\sigma_{\min}$ curves.

(b) Eigenvalue and eigenvector errors.

Figure 12: Left: $\sigma_{\min}(S(AQ_{20} - \lambda Q_{20}))$ across different λ. Right: The eigenvector angle (dashed) and refined eigenvalue error (solid) obtained via the sketched minimum perturbation extraction. The $\ell = r$ case is solved directly as a randomized Rayleigh-Ritz problem [36] and is included for reference.

As shown in Figure 12a, the continuous residual curves $\lambda \mapsto \sigma_{\min}(S(AQ - \lambda Q))$ maintain a sharp minimum near the exact eigenvalue across various sketch dimensions, seemingly preserving the global singularity condition. Moreover, Figure 12b confirms that the refined eigenvalue errors adhere to the expected theoretical second-order decay relative to the eigenvector angle. However, a critical bottleneck emerges. Even as the sketch dimension approaches the full original dimension ($l = n$), the sketched extraction does not reach the accuracy of the unsketched baseline.

While standard sketching techniques successfully preserve dominant singular subspaces for tasks such as randomized SVD [22], relying on global projection error bounds is insufficient here. The minimum perturbation method extracts approximate eigenvectors by solving a generalized eigenvalue problem restricted to the trailing right singular space $V_2 \in \mathbb{R}^{2r \times r}$ (associated with the r smallest singular values of $C = [Q, AQ]$). Consequently, the sketch should preserve the internal geometry of this trailing subspace.

Applying Corollary 2.1 to $C = [Q, AQ] \in \mathbb{R}^{n \times 2r}$ with trailing block size $k = r$ and using S as a γ-subspace embedding for range(C) yields

$$\|\sin\Theta(\tilde{V}_2, V_2)\|_2 \leq \frac{\gamma\sigma_r(C)\sigma_{r+1}(C)}{(1-\gamma)\sigma_r^2(C) - \sigma_{r+1}^2(C)}, \tag{23}$$

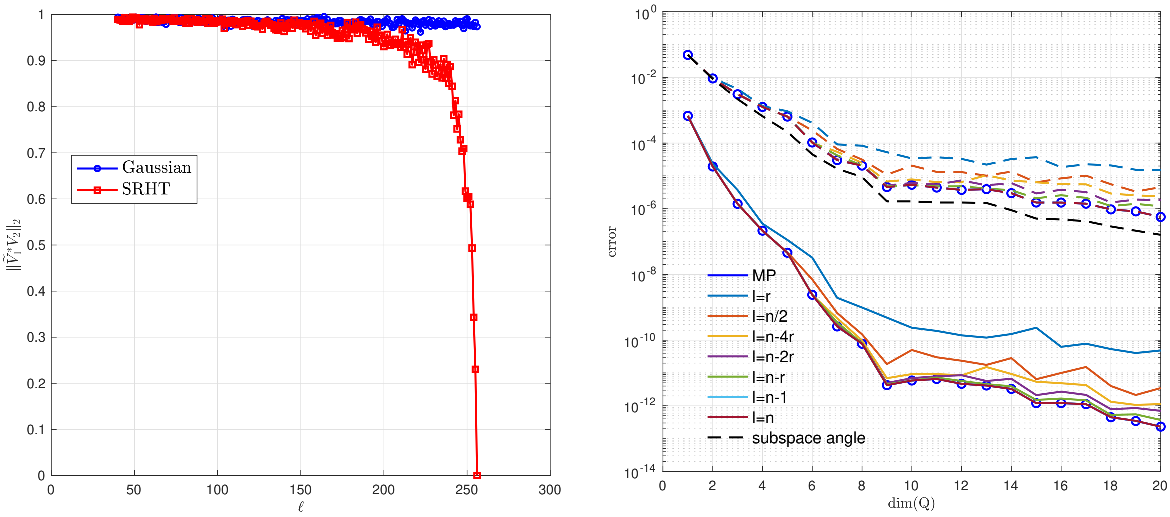
provided $(1-\gamma)\,\sigma_r^2(C) > \sigma_{r+1}^2(C)$. This bound governs the full trailing block and it can be large when the ratio $\sigma_r(C)/\sigma_{r+1}(C)$ is modest. By contrast, Lemma 4.1 shows that the target direction y_* can remain close to span$(\tilde{V}_2)$ without requiring the full trailing subspace $\tilde{V}_2$ to be close to V_2. Together, the two bounds suggest a distinction between detection and extraction. The containment estimate explains why the eigenvalue dip can survive sketching, whereas the subspace-rotation bound (23) indicates why accurate extraction may still fail when the geometry of the full trailing subspace is not well-preserved.

The explicit distortion bounds in Section 2.5 predict qualitatively different behavior for the two sketching distributions. The Gaussian distortion (5) has an irreducible floor of order $\sqrt{k/n}$ even at $\ell = n$, so (23) can never be driven to zero. The SRHT distortion (6) is proportional to $n - \ell$ and vanishes at $\ell = n$, so the trailing subspace $\tilde{V}_2$ can in principle recover V_2 when l is close to n.

Discussion. We verify this structural separation at $\dim(Q) = 20$ by monitoring the subspace rotation $\|\tilde{V}_1^\top V_2\|_2 = \|\sin\Theta(\tilde{V}_2, V_2)\|_2$ as the sketch dimension ℓ varies. Figure 13a shows that this quantity remains close to 1 for both sketches at most ℓ, confirming that the trailing subspace is genuinely scrambled. The SRHT curve begins to drop only when ℓ approaches n, and the Gaussian curve stays near 1 throughout. These are consistent with the distortion floor in (5).

We also verified that the containment angle $\sin\measuredangle(y_*, \mathrm{span}(\tilde{V}_2))$ remains between 4.5×10^{-7} and 6.5×10^{-7} for all l and both sketch types, comparable to the unsketched eigenvector accuracy $\measuredangle(Q, v) \approx 1.6 \times 10^{-7}$. This is consistent with Lemma 4.1. The target direction is not lost under sketching, even though the full trailing subspace may be strongly rotated.

Figure 13b repeats the extraction experiment of Figure 12b using SRHT sketches with



(a) $\|\sin\Theta(\tilde{V}_2, V_2)\|_2$.

(b) SRHT-sketched MP extraction.

Figure 13: Subspace diagnostics, $n = 256$. Left: $\|\sin\Theta(\tilde{V}_2, V_2)\|_2$ at $\dim(Q) = 20$ as a function of the sketch dimension ℓ. Right: SRHT-sketched MP extraction across increasing $\dim(Q)$ for $\ell \in \{2r, n/2, n-4r, n-2r, n-r, n-1, n\}$; eigenvalue errors (solid) and eigenvector angles (dashed).

$n = 256$. At small sketch dimensions the eigenvalue errors plateau well above the unsketched baseline, mirroring the Gaussian case. As ℓ increases toward n, the errors progressively approach the baseline, consistent with the vanishing SRHT distortion bound (6).

These experiments indicate that sketching the minimum perturbation extraction faces a structural obstacle. While the target eigenvector direction is preserved at all sketch dimensions, accurate extraction requires the full trailing subspace V_2 to be well-approximated, which demands sketch dimensions close to n. At such dimensions the computational savings over the unsketched method are marginal.

5 Conclusion

This thesis investigates the numerical solution of rectangular eigenvalue problems through three complementary experiments. The core computational pipeline combines rational approximation, linearization of finite representations, and minimum perturbation extraction. We designed these experiments to evaluate individual stages of this pipeline and to examine whether randomized sketching can reduce the cost of the extraction step.

For the Laplace eigenvalue problem in Section 4.1, direct scanning of $\sigma_{\min}(Q_B(\lambda))$ followed by local refinement proves to be the most reliable approach, and column-pivoted QR factorization with truncation stabilizes the basis construction. Since the single-eigenvalue scan fails to distinguish closely packed eigenvalues, we implemented a post-processing strategy to successfully resolve the eigenpairs within the cluster.

When applying the rational-approximation route, `sketchAAA` produces accurate surrogates of $Q_B(\lambda)$, and Procrustes alignment of the sampled bases further reduces the rational degree and improves approximation errors. However, the minimum perturbation extraction on the enlarged companion pencils is numerically ill-conditioned: the companion form is large and sparse, and the dense SVD at the core of the current MP method leads to a generalized eigenvalue problem with poorly conditioned matrices. At high collocation dimension, an additional difficulty arises before the approximation stage: $A(\lambda)$ becomes numerically rank-deficient, destroying the smoothness of the QR basis that the rational surrogate requires.

To determine whether this failure originates in the rational approximation or in the extraction machinery itself, Section 4.2 applies the pipeline to a quadratic rectangular problem arising from a damped wave equation. Because the collocation discretization directly yields a quadratic matrix-valued function, linearization produces a companion pencil of modest size without any surrogate construction. In this setting, the minimum perturbation method successfully identifies the physical eigenvalues while filtering out spurious candidates via a norm-aware residual criterion. By reusing the block structure of the pencil eigenvectors, the recovery step successfully avoids a full SVD at each candidate.

Section 4.3 examines whether randomized sketching can reduce the cost of the minimum perturbation extraction. Given a symmetric matrix A with a known eigenpair and a trial subspace range(Q), we applied sketching to the rectangular pencil $AQ - \lambda Q$. While the target direction remains well-contained in the sketched trailing subspace at all sketch dimensions, the eigenvector extraction accuracy does not reach the unsketched level at moderate dimensions. The numerical experiments relate this to the distortion of the trailing right singular subspace V_2. For the SRHT, the extraction errors progressively decrease and approach the unsketched baseline near $l \geq n - r$. The Gaussian sketch does not reach comparable accuracy at any tested sketch dimension. In both cases, the sketch dimensions required for accurate extraction are too large to offer meaningful cost savings.

Several directions remain open. A detailed investigation of error bounds on targeted eigenpairs solved via the minimum perturbation method is also worth pursuing. The companion pencils produced by linearization are large but highly structured and sparse, thus developing minimum perturbation solvers that exploit this block structure rather than performing a dense SVD could make the full pipeline practical at higher rational degrees. For the sketched extraction, the experiment in Section 4.3 suggests that applying sketching directly to the minimum perturbation method may not be ideal for preserving accuracy. More sophisticated randomized techniques that account for the structure of the extraction problem deserve further investigation.

References

[1] Pedro R. S. Antunes. "A well-conditioned method of fundamental solutions for Laplace equation". In: *Numer. Algorithms* 91.3 (2022), pp. 1381–1405. DOI: 10.1007/s11075-022-01306-x.

[2] A. Bermúdez et al. "Finite element analysis of a quadratic eigenvalue problem arising in dissipative acoustics". In: *SIAM J. Numer. Anal.* 38.1 (2000), pp. 267–291. DOI: 10.1137/S0036142999360160.

[3] Timo Betcke and Lloyd N. Trefethen. "Reviving the method of particular solutions". In: *SIAM Rev.* 47.3 (2005), pp. 469–491. DOI: 10.1137/S0036144503437336.

[4] Wolf-Jürgen Beyn. "An integral method for solving nonlinear eigenvalue problems". In: *Linear Algebra Appl.* 436.10 (2012), pp. 3839–3863. DOI: 10.1016/j.laa.2011.03.030.

[5] James Bisgard. *Analysis and linear algebra: the singular value decomposition and applications*. Vol. 94. Student Mathematical Library. American Mathematical Society, Providence, RI, [2021] ©2021, pp. xviii+217.

[6] Gregory Boutry et al. "The generalized eigenvalue problem for nonsquare pencils using a minimal perturbation approach". In: *SIAM J. Matrix Anal. Appl.* 27.2 (2005), pp. 582–601. DOI: 10.1137/S0895479803428795.

[7] Michael C. Brennan, Mark Embree, and Serkan Gugercin. "Contour integral methods for nonlinear eigenvalue problems: a systems theoretic approach". In: *SIAM Rev.* 65.2 (2023), pp. 439–470. DOI: 10.1137/20M1389303.

[8] Delin Chu and Gene H. Golub. "On a generalized eigenvalue problem for nonsquare pencils". In: *SIAM J. Matrix Anal. Appl.* 28.3 (2006), pp. 770–787. DOI: 10.1137/050628258.

[9] R. Courant and D. Hilbert. *Methods of mathematical physics. Vol. II: Partial differential equations*. (Vol. II by R. Courant.) Interscience Publishers (a division of John Wiley & Sons, Inc.), New York-London, 1962, pp. xxii+830.

[10] Saptarshi Das and Arnold Neumaier. "Solving overdetermined eigenvalue problems". In: *SIAM J. Sci. Comput.* 35.2 (2013), A541–A560. DOI: 10.1137/110828514.

[11] Chandler Davis and W. M. Kahan. "The rotation of eigenvectors by a perturbation. III". In: *SIAM J. Numer. Anal.* 7 (1970), pp. 1–46. DOI: 10.1137/0707001.

[12] James Weldon Demmel and Bo Kågström. "Computing stable eigendecompositions of matrix pencils". In: *Linear Algebra Appl.* 88/89 (1987), pp. 139–186. DOI: 10.1016/0024-3795(87)90108-X.

[13] Luca Dieci and Timo Eirola. "On smooth decompositions of matrices". In: *SIAM J. Matrix Anal. Appl.* 20.3 (1999), pp. 800–819. DOI: 10.1137/S0895479897330182.

[14] Tobin A. Driscoll. "Eigenmodes of isospectral drums". In: *SIAM Rev.* 39.1 (1997), pp. 1–17. DOI: 10.1137/S0036144595285069.

[15] C. Effenberger. “Robust successive computation of eigenpairs for nonlinear eigenvalue problems”. In: *SIAM J. Matrix Anal. Appl.* 34.3 (2013), pp. 1231–1256. DOI: 10.1137/120885644.

[16] Steven Elsworth and Stefan Güttel. “Conversions between barycentric, RKFUN, and Newton representations of rational interpolants”. In: *Linear Algebra Appl.* 576 (2019), pp. 246–257. DOI: 10.1016/j.laa.2018.10.003.

[17] Brendan Gavin, Agnieszka Mi edlar, and Eric Polizzi. “FEAST eigensolver for nonlinear eigenvalue problems”. In: *J. Comput. Sci.* 27 (2018), pp. 107–117. DOI: 10.1016/j.jocs.2018.05.006.

[18] Gene H. Golub and Charles F. Van Loan. *Matrix computations*. Fourth. Johns Hopkins Studies in the Mathematical Sciences. Johns Hopkins University Press, Baltimore, MD, 2013, pp. xiv+756.

[19] Stefan Güttel, Daniel Kressner, and Bart Vandereycken. “Randomized sketching of nonlinear eigenvalue problems”. In: *SIAM J. Sci. Comput.* 46.5 (2024), A3022–A3043. DOI: 10.1137/22M153656X.

[20] Stefan Güttel, Gian Maria Negri Porzio, and Françoise Tisseur. “Robust rational approximations of nonlinear eigenvalue problems”. In: *SIAM J. Sci. Comput.* 44.4 (2022), A2439–A2463. DOI: 10.1137/20M1380533.

[21] Stefan Güttel and Françoise Tisseur. “The nonlinear eigenvalue problem”. In: *Acta Numer.* 26 (2017), pp. 1–94. DOI: 10.1017/S0962492917000034.

[22] N. Halko, P. G. Martinsson, and J. A. Tropp. “Finding structure with randomness: probabilistic algorithms for constructing approximate matrix decompositions”. In: *SIAM Rev.* 53.2 (2011), pp. 217–288. DOI: 10.1137/090771806.

[23] Behnam Hashemi, Yuji Nakatsukasa, and Lloyd N. Trefethen. “Rectangular eigenvalue problems”. In: *Adv. Comput. Math.* 48.6 (2022), Paper No. 80, 16. ISSN: 1019-7168,1572-9044. DOI: 10.1007/s10444-022-09994-8.

[24] Michiel E. Hochstenbach, Christian Mehl, and Bor Plestenjak. “Solving singular generalized eigenvalue problems by a rank-completing perturbation”. In: *SIAM J. Matrix Anal. Appl.* 40.3 (2019), pp. 1022–1046. DOI: 10.1137/18M1188628.

[25] Shinji Ito and Kazuo Murota. “An algorithm for the generalized eigenvalue problem for nonsquare matrix pencils by minimal perturbation approach”. In: *SIAM J. Matrix Anal. Appl.* 37.1 (2016), pp. 409–419. DOI: 10.1137/14099231X.

[26] Peter Kunkel and Volker Mehrmann. “Smooth factorizations of matrix valued functions and their derivatives”. In: *Numer. Math.* 60.1 (1991), pp. 115–131. DOI: 10.1007/BF01385717.

[27] R. Sherman Lehman. “Developments at an analytic corner of solutions of elliptic partial differential equations”. In: *J. Math. Mech.* 8 (1959), pp. 727–760. DOI: 10.1512/iumj.1959.8.58047.

[28] Jiao-fen Li et al. “A Riemannian optimization approach for solving the generalized eigenvalue problem for nonsquare matrix pencils”. In: *J. Sci. Comput.* 82.3 (2020), Paper No. 67, 43. DOI: 10.1007/s10915-020-01173-5.

[29] Ren-Cang Li. “Relative perturbation theory. II. Eigenspace and singular subspace variations”. In: *SIAM J. Matrix Anal. Appl.* 20.2 (1999), pp. 471–492. DOI: 10.1137/S0895479896298506.

[30] Pieter Lietaert et al. “Automatic rational approximation and linearization of nonlinear eigenvalue problems”. In: *IMA J. Numer. Anal.* 42.2 (2022), pp. 1087–1115. DOI: 10.1093/imanum/draa098.

[31] D. Steven Mackey et al. “Vector spaces of linearizations for matrix polynomials”. In: *SIAM J. Matrix Anal. Appl.* 28.4 (2006), pp. 971–1004. DOI: 10.1137/050628350.

[32] C. B. Moler and G. W. Stewart. “An algorithm for generalized matrix eigenvalue problems”. In: *SIAM J. Numer. Anal.* 10 (1973), pp. 241–256. DOI: 10.1137/0710024.

[33] Yuji Nakatsukasa, Olivier Sète, and Lloyd N. Trefethen. “The AAA algorithm for rational approximation”. In: *SIAM J. Sci. Comput.* 40.3 (2018), A1494–A1522. DOI: 10.1137/16M1106122.

[34] Taejun Park and Yuji Nakatsukasa. “A fast randomized algorithm for computing an approximate null space”. In: *BIT* 63.2 (2023), Paper No. 36, 27. DOI: 10.1007/s10543-023-00979-7.

[35] Mark Rudelson and Roman Vershynin. “Non-asymptotic theory of random matrices: extreme singular values”. In: *Proceedings of the International Congress of Mathematicians. Volume III*. Hindustan Book Agency, New Delhi, 2010, pp. 1576–1602.

[36] Nian Shao. “Stabilizing the Rayleigh–Ritz procedure by randomization”. In: *arXiv preprint arXiv:2604.01037* (2026). DOI: 10.48550/arXiv.2604.01037.

[37] Joel A. Tropp. “Improved analysis of the subsampled randomized Hadamard transform”. In: *Adv. Adapt. Data Anal.* 3.1-2 (2011), pp. 115–126. DOI: 10.1142/S1793536911000787.

[38] Sabine Van Huffel and Joos Vandewalle. *The total least squares problem*. Vol. 9. Frontiers in Applied Mathematics. Computational aspects and analysis, With a foreword by Gene H. Golub. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 1991, pp. xiv+300. DOI: 10.1137/1.9781611971002.

[39] David P. Woodruff. “Sketching as a tool for numerical linear algebra”. In: *Found. Trends Theor. Comput. Sci.* 10.1-2 (2014), pp. iv+157. DOI: 10.1561/0400000060.